

Engineering Mathematics

Previous Year Questions • 1995–2026

CALCULUS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Calculus (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
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- 5+ years of teaching experience



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- AIR 19 — GATE (IN)
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- 3+ years in teaching & the VLSI industry



> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the questions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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Calculus

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7. Equation of Tangent and Normal
8. Monotonicity, Maxima and Minima
9. Curvature and Inflection Points
10. Mean Value Theorems and Taylor Series
11. Partial Derivatives
12. Indefinite Integration
13. Definite Integration
14. Area Under a Curve and Curve Tracing
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16. Order and Region of Integration
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18. Improper Integrals and Limit of Sum

Branches: EC,EE,IN,CS,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 54 **MCQ:** 34 **MSQ:** 2 **NAT:** 18
1M: 22 **2M:** 32 **Desc:** 0 **Avg Weightage** ~4M **Range:** 2M(2023)–5M(2018)

EE — Electrical Engineering

Questions: 35 **MCQ:** 24 **MSQ:** 0 **NAT:** 11
1M: 18 **2M:** 17 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–4M(2018)

IN — Instrumentation Engineering

Questions: 32 **MCQ:** 23 **MSQ:** 1 **NAT:** 8
1M: 18 **2M:** 14 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2022)–4M(2020)

CS — Computer Science & IT

Questions: 38 **MCQ:** 24 **MSQ:** 3 **NAT:** 11
1M: 29 **2M:** 9 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2025)–4M(2017)

ME — Mechanical Engineering

Questions: 48 **MCQ:** 42 **MSQ:** 0 **NAT:** 6
1M: 28 **2M:** 20 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–6M(2010)

CE — Civil Engineering

Questions: 57 **MCQ:** 48 **MSQ:** 2 **NAT:** 7
1M: 34 **2M:** 23 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–12M(2016)

CH — Chemical Engineering

Questions: 28 **MCQ:** 24 **MSQ:** 1 **NAT:** 3
1M: 16 **2M:** 12 **Desc:** 0 **Avg Weightage** ~2M **Range:** 1M(2026)–4M(2022)

PI — Production & Industrial Engineering

Questions: 26 **MCQ:** 19 **MSQ:** 0 **NAT:** 7
1M: 18 **2M:** 8 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–6M(2016)

**See preface for how Avg Weightage and Range are calculated.*

Electronics & Communication Engineering

Q2.26.1 NAT 2026 2M

Consider the square region R in the X-Y plane as shown with the dark shading in the Figure. The value of $\iint_R (x^2 + y^2 - 1)\,dx\,dy$ is _____.
(rounded off to two decimal places)

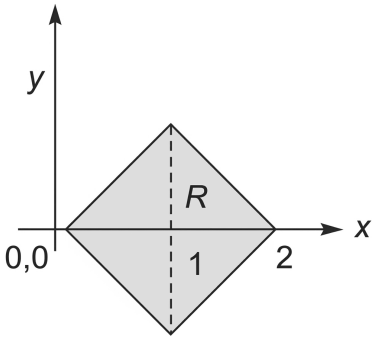


Fig. Q2.26.1

Q2.26.2 MCQ 2026 2M

Consider the two series, S_A and S_B, where

$$S_A = \sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

$$S_B = 1 + \frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} + \frac{1}{128} + \frac{1}{512} + \cdots$$

Which of the following statements is correct for the two given series?

A. Both S_A and S_B converge.

B. Neither S_A nor S_B converges.

C. S_A converges but S_B does not converge.

D. S_B converges but S_A does not converge.

Q2.25.1 MSQ 2025 1M

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as

$$f(x) = 2x^3 - 3x^2 - 12x + 1$$

Which of the following statements is/are correct? (Here, $\mathbb{R}$ is the set of real numbers.)

☐ **A.** f has no global maximizer

☐ **B.** f has no global minimizer

☐ **C.** $x = -1$ is a local minimizer of f

☐ **D.** $x = 2$ is a local maximizer of f

Q2.25.2 MCQ 2025 1M

Consider the following series:

(i) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

(ii) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

(iii) $\sum_{n=1}^{\infty} \frac{1}{n!}$

Choose the correct option.

A. Only (ii) converges

B. Only (ii) and (iii) converge

C. Only (iii) converges

D. All three converge

Q2.25.3 MSQ 2025 2M

Consider a non-negative function $f(x)$ which is continuous and bounded over the interval $[2, 8]$. Let M and m denote, respectively, the maximum and the minimum values of $f(x)$ over the interval.
Among the combinations of α and β given below, choose the one(s) for which the inequality

$$\beta \le \int_2^8 f(x)\,dx \le \alpha$$

is guaranteed to hold.

☐ **A.** $\beta = 5m,\ \alpha = 7M$

☐ **B.** $\beta = 6m,\ \alpha = 5M$

☐ **C.** $\beta = 7m,\ \alpha = 6M$

☐ **D.** $\beta = 7m,\ \alpha = 5M$

Q2.23.1 MCQ 2023 2M

Let $v_1 = \begin{bmatrix}1\\2\\0\end{bmatrix}$ and $v_2 = \begin{bmatrix}2\\1\\3\end{bmatrix}$ be two vectors. The value of the coefficient α in the expression $v_1 = \alpha v_2 + e$, which minimizes the length of the error vector e, is

A. $\frac{7}{2}$ **B.** $\frac{-2}{7}$ **C.** $\frac{2}{7}$ **D.** $\frac{-7}{2}$

Q2.22.1 MCQ 2022 2M

The function $f(x) = 8\log_e x - x^2 + 3$ attains its minimum over the interval $[1, e]$ at $x =$ _____. (Here $\log_e x$ is the natural logarithm of x.)

A. 2 **B.** 1

C. e **D.** $\frac{1+e}{2}$

Q2.22.2 NAT 2022 2M

The value of the integral $\iint_D 3(x^2+y^2)dxdy$, where D is the shaded triangular region shown in the diagram, is _____ (rounded off to the nearest integer).

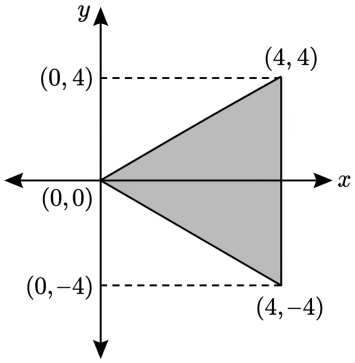


Fig. Q2.22.2

Q2.20.1 NAT 2020 2M

For the solid S shown below, the value of $\iiint_S xdxdydz$ (rounded off to two decimal places) is _____

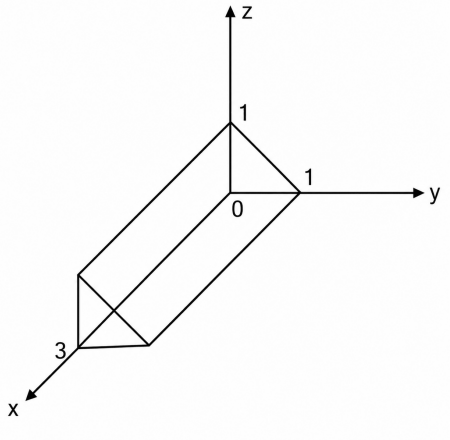


Fig. Q2.20.1

Q2.20.2 MCQ 2020 1M

The partial derivative of the following function $f(x, y, z) = e^{1-x\cos y} + xze^{-\frac{1}{(1+y^2)}}$ With respect to x at the point $(1, 0, e)$ is

A. 1 **B.** 0

C. $\frac{1}{e}$ **D.** -1

Q2.19.1 NAT 2019 2M

The value of the integral $\int_0^{\pi}\int_y^{\pi}\frac{\sin x}{x}dxdy$, is equal to _____.

Q2.19.2 MCQ 2019 2M

Consider a differentiable function $f(x)$ on the set of real numbers such that $f(-1) = 0$ and $|f'(x)| \le 2$. Given these conditions, which one of the following inequalities is necessarily true for all $x \in [-2, 2]$?

A. $f(x) \le \frac{1}{2}|x+1|$ **B.** $f(x) \le \frac{1}{2}|x|$

C. $f(x) \le 2|x+1|$ **D.** $f(x) \le 2|x|$

Q2.18.1 MCQ 2018 2M

Let $f(x, y) = \frac{ax^2+by^2}{xy}$, where a and b are constants. If $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$ at $x = 1$ and $y = 2$, then the relation between a and b is

A. $a = \frac{b}{4}$ **B.** $a = \frac{b}{2}$

C. a = 2b **D.** a = 4b

Q2.18.2 NAT 2018 1M

Taylor series expansion of $f(x) = \int_0^x e^{-\left(\frac{t^2}{2}\right)} dt$ around $x = 0$ has the form $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots$. The coefficient a_2 (correct to two decimal places) is equal to _____

Q2.18.3 NAT 2018 2M

Let $r = x^2 + y - z$ and $z^3 - xy + yz + y^3 = 1$. Assume that x and y are independent variables. At $(x, y, z) = (2, -1, 1)$, the value (correct to two decimal places) of $\frac{\partial r}{\partial x}$ is _____

Q2.17.1 NAT 2017 2M

The minimum value of the function $f(x) = \frac{1}{3}x(x^2 - 3)$ in the interval $-100 \le x \le 100$ occurs at $x =$ _____.

Q2.17.2 MCQ 2017 2M

The values of the integrals $\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dy\right) dx$ and $\int_0^1 \left(\int_0^1 \frac{x-y}{(x+y)^3} dx\right) dy$ are

A. same and equal to 0.5 **B.** same and equal to -0.5

C. 0.5 and -0.5 respectively **D.** -0.5 and 0.5 respectively

Q2.17.3 MCQ 2017 2M

Let $f(x) = e^{x+x^2}$ for real x. From among the following, choose the Taylor series approximation of $f(x)$ around $x = 0$, which includes all powers of x less than or equal to 3.

A. $1 + x + x^2 + x^3$ **B.** $1 + x + \frac{3}{2}x^2 + x^3$

C. $1 + x + \frac{3}{2}x^2 + \frac{7}{6}x^3$ **D.** $1 + x + 3x^2 + 7x^3$

Q2.17.4 NAT 2017 2M

A three dimensional region R of the finite volume is described by $x^2 + y^2 \le z^3$, $0 \le z \le 1$. Where x, y, z are real. The volume of R (correct to two decimal places) is _____.

Q2.16.1 NAT 2016 1M

The integral $\frac{1}{2\pi} \iint_D (x + y + 10) dx dy$ where D denotes the disc: $x^2 + y^2 \le 4$, evaluates to _____.

Q2.16.2 MCQ 2016 1M

As x varies from -1 to 3, which of the following describes the behaviour of the function $f(x) = x^3 - 3x^2 + 1$?

A. $f(x)$ increases monotonically

B. $f(x)$ increases, then decreases and increases again

C. $f(x)$ decreases, then increases and decreases again

D. $f(x)$ increases and then decreases

Q2.16.3 NAT 2016 2M

The region specified by $\{(\rho, \phi, z) : 3 \le \rho \le 5, \quad \frac{\pi}{8} \le \phi \le \frac{\pi}{4}, \quad 3 \le z \le 4.5\}$ in cylindrical coordinates has volume of _____.

Q2.16.4 MCQ 2016 1M

Given the following statements about a function $f : \mathbb{R} \to \mathbb{R}$, select the right option:

P: If $f(x)$ is continuous at $x = x_0$, then it is also differentiable at $x = x_0$
Q: If $f(x)$ is continuous at $x = x_0$, then it may not be differentiable at $x = x_0$
R: If $f(x)$ is differentiable at $x = x_0$, then it is also continuous at $x = x_0$

A. P is true, Q is false, R is false

B. P is false, Q is true, R is true

C. P is false, Q is true, R is false

D. P is true, Q is false, R is true

Q2.16.5 NAT 2016 2M

The integral $\int_0^1 \frac{dx}{\sqrt{(1-x)}}$ is equal to _____.

Q2.16.6 MCQ 2016 1M

How many distinct values of x satisfy the equation $\sin(x) = x/2$, where x is in radians?

A. 1 **B.** 2 **C.** 3 **D.** 4 or more

Q2.16.7 NAT 2016 1M

A triangle in the xy-plane is bounded by the straight lines $2x = 3y$, $y = 0$ and $x = 3$. The volume above the triangle and under the plane $x + y + z = 6$ is _____.

Q2.15.1 NAT 2015 2M

The maximum area (in sqare units) of a rectangle whose vertices lie on the ellipse $x^2 + 4y^2 = 1$ is _____.

Q2.15.2 NAT 2015 1M

The value of the integral $\int_{-\infty}^{\infty} 12\cos(2\pi t)\dfrac{\sin(4\pi t)}{4\pi t}dt$ is _____.

Q2.15.3 MCQ 2015 1M

A function $f(x) = 1 - x^2 + x^3$ is defined in the closed interval $[-1, 1]$. The value of x, in the open interval $(-1, 1)$ for which the mean value theorem is satisfied, is

A. $-\dfrac{1}{2}$ **B.** $-\dfrac{1}{3}$ **C.** $\dfrac{1}{3}$ **D.** $\dfrac{1}{2}$

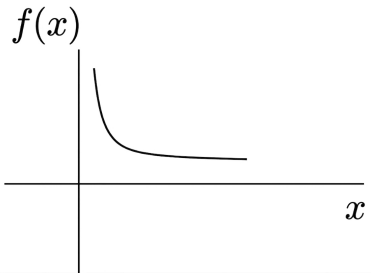
Q2.15.4 NAT 2015 1M

The value of $\sum_{n=0}^{\infty} n\left(\dfrac{1}{2}\right)^n$ is _____.

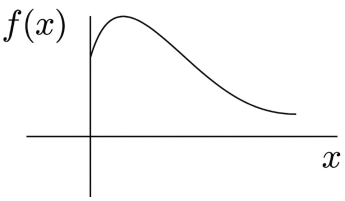
Q2.15.5 MCQ 2015 1M

Which one of the following graphs describes the following function? $f(x) = e^{-x}(x^2 + x + 1)$?

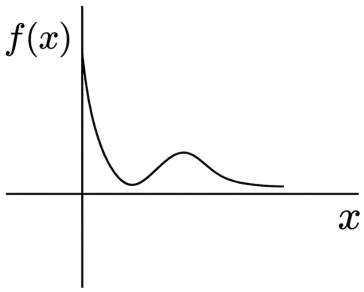
A.



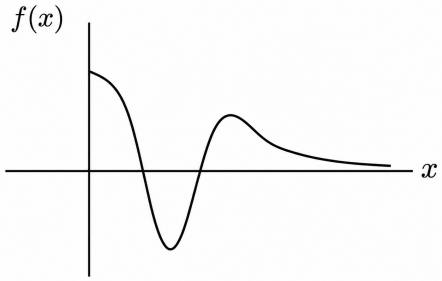
B.



C.



D.



Q2.15.6 MCQ 2015 1M

The contour on the $x - y$ plane, where the partial derivative of $x^2 + y^2$ with respect to y is equal to the partial derivative of $6y + 4x$ with respect to 'x', is

A. $y = 2$ **B.** $x = 2$

C. $x + y = 4$ **D.** $x - y = 0$

Q2.14.1 MCQ 2014 2M

The Taylor series expansion of $3\sin x + 2\cos x$ is

A. $2 + 3x - x^2 - \dfrac{x^3}{2} + \ldots$

B. $2 - 3x + x^2 - \dfrac{x^3}{2} + \ldots$

C. $2 + 3x + x^2 + \dfrac{x^3}{2} + \ldots$

D. $2 - 3x - x^2 + \dfrac{x^3}{2} + \ldots$

Q2.14.2 NAT 2014 2M

The volume under the surface $z(x, y) = x + y$ and above the triangle in the xy plane defined by $\{0 \le y \le x$ and $0 \le x \le 12\}$ is _____.

Q2.14.3 MCQ 2014 1M

For $0 \le t < \infty$, the maximum value of the function $f(t) = e^{-t} - 2e^{-2t}$ occurs at

A. $t = \log_e 4$ **B.** $t = \log_e 2$

C. $t = 0$ **D.** $t = \log_e 8$

Q2.14.4 MCQ 2014 1M

The value of $\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^x$ is

A. $\ln 2$ **B.** 1.0 **C.** e **D.** ∞

Q2.14.5 NAT 2014 1M

The maximum value of the function $f(x) = \ln(1 + x) - x$ (where $x > -1$) occurs at $x =$ _____.

Q2.14.6 MCQ 2014 1M

The series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges to

A. $2 \ln 2$ **B.** $\sqrt{2}$ **C.** 2 **D.** e

Q2.14.7 NAT 2014 2M

The maximum value of $f(x) = 2x^3 - 9x^2 + 12x - 3$ in the interval $0 \le x \le 3$ is _____.

Q2.14.8 MCQ 2014 2M

For a right angled triangle, if the sum of the lengths of the hypotenuse and a side is kept constant, in order to have maximum area of the triangle, the angle between the hypotenuse and the side is

A. 12° **B.** 36° **C.** 60° **D.** 45°

Q2.14.9 MCQ 2014 1M

If $z = xy \ln(xy)$, then

A. $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 0$ **B.** $y\frac{\partial z}{\partial x} = x\frac{\partial z}{\partial y}$

C. $x\frac{\partial z}{\partial x} = y\frac{\partial z}{\partial y}$ **D.** $y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} = 0$

Q2.12.1 MCQ 2012 2M

The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is

A. 21 **B.** 25 **C.** 41 **D.** 46

Q2.10.1 MCQ 2010 2M

If $e^y = x^{1/x}$ then y has a

A. maximum at $x = e$

B. minimum at $x = e$

C. maximum at $x = e^{-1}$

D. minimum at $x = e^{-1}$

Q2.09.1 MCQ 2009 2M

The Taylor series expansion of $\frac{\sin x}{x - \pi}$ at $x = \pi$ is given by

A. $1 + \frac{(x-\pi)^2}{3!} + \ldots$

B. $-1 - \frac{(x-\pi)^2}{3!} + \ldots$

C. $1 - \frac{(x-\pi)^2}{3!} + \ldots$

D. $-1 + \frac{(x-\pi)^2}{3!} + \ldots$

Q2.08.1 MCQ 2008 1M

For real values of x, the minimum value of function $f(x) = e^x + e^{-x}$ is

A. 2 **B.** 1 **C.** 0.5 **D.** 0

Q2.08.2 MCQ 2008 1M

Which of the following function would have only odd powers of x in its Taylor series expansion about the point $x = 0$?

A. $\sin(x^3)$ **B.** $\sin(x^2)$

C. $\cos(x^3)$ **D.** $\cos(x^2)$

Q2.08.3 MCQ 2008 2M

In the Taylor series expansion of $e^x + \sin x$ about the point $x = \pi$, the coefficient of $(x - \pi)^2$ is

A. e^π **B.** $0.5e^\pi$

C. $e^\pi + 1$ **D.** $e^\pi - 1$

Q2.08.4 MCQ 2008 2M

The value of the integral of the function $g(x, y) = 4x^3 + 10y^4$ along the straight line segment from the point $(0, 0)$ to the point $(1, 2)$ in the xy-plane is

A. 33 **B.** 35 **C.** 40 **D.** 56

Q2.07.1 MCQ 2007 1M

$\lim_{\theta \to 0} \frac{\sin(\theta/2)}{\theta}$ is

A. 0.5 **B.** 1

C. 2 **D.** $not defined$

Q2.07.2 MCQ 2007 2M

The following plot shows a function y which varies linearly with x. The value of the integral $I = \int_1^2 y\,dx$ is

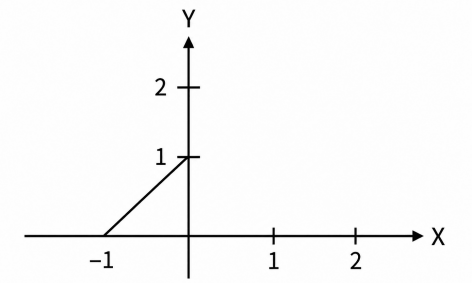


Fig. Q2.07.2

A. 1 **B.** 2.5

C. 4 **D.** 5

Q2.07.3 MCQ 2007 2M

For the function e^{-x}, the linear approximation around $x = 2$ is

A. $(3 - x)e^{-2}$

B. $1 - x$

C. $\left[3 + 2\sqrt{2} - (1 + \sqrt{2})x\right] e^{-2}$

D. e^{-2}

Q2.07.4 MCQ 2007 2M

For $|x| \ll 1$, $\coth(x)$ can be approximated as

A. x **B.** x^2

C. $\frac{1}{x}$ **D.** $\frac{1}{x^2}$

Q2.07.5 MCQ 2007 2M

Consider the function $f(x) = x^2 - x - 2$. The maximum value of $f(x)$ in the closed interval $[-4, 4]$ is

A. 18 **B.** 10

C. −2.25 **D.** indeterminate

Q2.05.1 MCQ 2005 2M

The value of the integral $I = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-x^2/8}\,dx$ is _____.

A. 1 **B.** π **C.** 2 **D.** 2π

Electrical Engineering

Q2.26.1 NAT 2026 2M

The integral $\frac{1}{\pi} \int_0^\infty \frac{x^{2026}}{(1 + x^{2026})(1 + x^2)}\,dx$ evaluates to _____.

(Round off to two decimal places)

Q2.25.1 MCQ 2025 1M

Consider the set S of points $(x, y) \in \mathbb{R}^2$ which minimize the real valued function

$$f(x, y) = (x + y - 1)^2 + (x + y)^2$$

Which of the following statements is true about the set S?

A. The number of elements in the set S is finite and more than one.

B. The number of elements in the set S is infinite.

C. The set S is empty.

D. The number of elements in the set S is exactly one.

Q2.22.1 MCQ 2022 2M

Let $f(x) = \int_0^x e^t(t - 1)(t - 2)dt$. Then $f(x)$ decreases in the interval

A. $x \in (1, 2)$ **B.** $x \in (2, 3)$

C. $x \in (0, 1)$ **D.** $x \in (0.5, 1)$

Q2.21.1 NAT 2021 1M

Suppose the circle $x^2 + y^2 = 1$ and $(x - 1)^2 + (y - 1)^2 = r^2$ intersect each other orthogonally at the point (u, v). Then $u + v =$ _____.

Q2.21.2 MCQ 2021 1M

Let $f(x)$ be a real-valued function such that $f'(x_0) = 0$ for some $x_0 \in (0, 1)$, and $f''(x) > 0$ for all $x \in (0, 1)$. Then $f(x)$ has

A. exactly one local minimum in $(0, 1)$

B. one local maximum in $(0, 1)$

C. no local minimum in $(0, 1)$

D. two distinct local minimum in $(0, 1)$

Q2.20.1 MCQ 2020 1M

Which of the following is true for all possible non-zero choices of integers m, n; $m \neq n$, or all possible non-zero choices of real numbers p, q; $p \neq q$ as applicable?

A. $\frac{1}{\pi}\int_0^{\pi} \sin m\theta \sin(n\theta)d\theta = 0$

B. $\frac{1}{2\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin(p\theta)\sin(q\theta)d\theta = 0$

C. $\lim_{\alpha\to\infty} \frac{1}{2\alpha}\int_{-\alpha}^{\alpha} \sin p\theta \sin q\theta d\theta = 0$

D. $\frac{1}{2\pi}\int_{-\pi}^{\pi} \sin(p\theta)\cos(q\theta)d\theta = 0$

Q2.20.2 MCQ 2020 2M

For real number x and y with $y = 3x^2+3x+1$, the maximum and minimum value of y for $x \in [-2, 0]$ are respectively

A. 1 and $\frac{1}{4}$ **B.** -2 and $\frac{-1}{2}$

C. 7 and $\frac{1}{4}$ **D.** 7 and 1

Q2.18.1 MCQ 2018 1M

Let f be a real-valued function of a real variable defined as $f(x) = x^2$ for $x \geq 0$, and $f(x) = -x^2$ for $x < 0$. Which one of the following statements is true?

A. $f(x)$ is discontinuous at $x = 0$

B. $f(x)$ is continuous but not differentiable at $x = 0$

C. $f(x)$ is differentiable but its first derivative is not continuous at $x = 0$

D. $f(x)$ is differentiable but its first derivative is not differentiable at $x = 0$

Q2.18.2 NAT 2018 2M

Let $f(x) = 3x^3 - 7x^2 + 5x + 6$. The maximum value of $f(x)$ over the interval $[0, 2]$ is _____ (up to 1 decimal place).

Q2.18.3 NAT 2018 1M

Let f be a real-valued function of a real variable defined as $f(x) = x - [x]$, where $[x]$ denotes the largest integer less than or equal to x. The value of $\int_{0.25}^{1.25} f(x)dx$ is _____ (up to 2 decimal places).

Q2.17.1 NAT 2017 1M

Let $I = c\iint_R xy^2dxdy$, where R is the region shown in the figure and $c = 6 \times 10^{-4}$. The value of I equals _____. (Give the answer up to two decimal places)

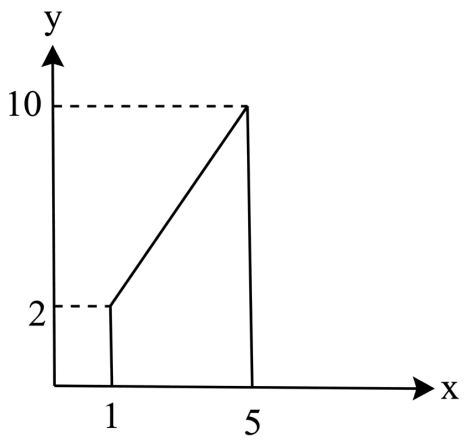


Fig. Q2.17.1

Q2.17.2 MCQ 2017 2M

A function $f(x)$ is defined as

$$f(x) = \begin{cases} e^x & ,x < 1 \\ \ln x + ax^2 + bx & ,x \geq 1 \end{cases} \text{ where } x \in \mathbb{R}.$$

Which one of the following statements is TRUE?

A. $f(x)$ is NOT differentiable at $x = 1$ for any values of a and b.

B. $f(x)$ is differentiable at $x = 1$ for the unique values of a and b.

C. $f(x)$ is differentiable at $x = 1$ for all values of a and b such that $a + b = e$.

D. $f(x)$ is differentiable at $x = 1$ for all values of a and b.

Q2.17.3 NAT 2017 1M

Let $y^2 - 2y + 1 = x$ and $\sqrt{x} + y = 5$. The value of $x + \sqrt{y}$ equals _____. (Given the answer up to three decimal places)

Q2.17.4 NAT 2017 1M

Consider a function $f(x, y, z)$ given by $f(x, y, z) = (x^2 + y^2 - 2z^2)(y^2 + z^2)$. The partial derivative of this function with respect to x at the point $x = 2, y = 1$ and $z = 3$ is _____.

Q2.17.5 NAT 2017 1M

Let x and y be integers satisfying the following equations

$$2x^2 + y^2 = 34$$
$$x + 2y = 11$$

The value of $(x + y)$ is _____.

Q2.17.6 MCQ 2017 2M

Let $g(x) = \begin{cases} -x, & x \le 1 \\ x + 1 & x \ge 1 \end{cases}$ and $f(x) = \begin{cases} 1 - x, & x \le 0 \\ x^2, & x > 0 \end{cases}$.
Consider the composition of f and g, i.e., $(f \circ g)(x) = f(g(x))$. The number of discontinuities in $(f \circ g)(x)$ present in the interval $(-\infty, 0)$ is

A. 0 **B.** 1
C. 2 **D.** 4

Q2.16.1 NAT 2016 2M

Let $S = \sum_{n=0}^{\infty} n\alpha^n$ where $|\alpha| < 1$. The value of α in the range $0 < \alpha < 1$, such that $S = 2\alpha$ is _____.

Q2.16.2 MCQ 2016 2M

The value of the integral $2\int_{-\infty}^{\infty} \left(\frac{\sin 2\pi t}{\pi t}\right) dt$ is equal to

A. 0 **B.** 0.5
C. 1 **D.** 2

Q2.16.3 NAT 2016 1M

The maximum value attained by the function $f(x) = x(x - 1)(x - 2)$ in the interval $[1, 2]$ is _____.

Q2.14.1 MCQ 2014 2M

Let $f(x) = xe^{-x}$. The maximum value of the function in the interval $(0, \infty)$ is

A. e^{-1} **B.** e
C. $1 - e^{-1}$ **D.** $1 + e^{-1}$

Q2.14.2 MCQ 2014 1M

Minimum of the real valued function $f(x) = (x - 1)^{2/3}$ occurs at x equal to

A. $-\infty$ **B.** 0 **C.** 1 **D.** ∞

Q2.14.3 MCQ 2014 2M

To evaluate the double integral $\int_0^8 \left(\int_{y/2}^{(y/2)+1} \left(\frac{2x - y}{2}\right) dx\right) dy$, we make the substitution $u = \left(\frac{2x - y}{2}\right)$ and $v = \frac{y}{2}$. The integral will reduce to

A. $\int_0^4 \left(\int_0^2 2u\,du\right) dv$

B. $\int_0^4 \left(\int_0^1 2u\,du\right) dv$

C. $\int_0^4 \left(\int_0^1 u\,du\right) dv$

D. $\int_0^4 \left(\int_0^2 u\,du\right) dv$

Q2.14.4 MCQ 2014 2M

The minimum value of the function $f(x) = x^3 - 3x^2 - 24x + 100$ in the interval $[-3, 3]$ is

A. 20 **B.** 28 **C.** 16 **D.** 32

Q2.14.5 NAT 2014 1M

A particle, starting from origin at $t = 0$ s, is traveling along x – axis with velocity

$$v = \frac{\pi}{2}\cos\left(\frac{\pi}{2}t\right) \text{ m/s}$$

At $t = 3$ s, the difference between the distance covered by the particle and the magnitude of displacement from the origin is _____.

Q2.13.1 MCQ 2013 2M

A function $y = 5x^2 + 10x$ is defined over an open interval $x = (1, 2)$. At least at one point in this interval, $\frac{dy}{dx}$ is exactly

A. 20 **B.** 25 **C.** 30 **D.** 35

Q2.11.1 MCQ 2011 1M

Roots of the algebraic equation $x^3 + x^2 + x + 1 = 0$ are

A. $(1, j, -j)$ **B.** $(1, -1, 1)$

C. $(0, 0, 0)$ **D.** $(-1, j, -j)$

Q2.11.2 MCQ 2011 2M

The function $f(x) = 2x - x^2 + 3$ has

A. a maxima at $x = 1$ and a minima at $x = 5$

B. a maxima at $x = 1$ and a minima at $x = -5$

C. only a maxima at $x = 1$

D. only a minima at $x = 1$

Q2.10.1 MCQ 2010 2M

At $t = 0$, the function $f(t) = \frac{\sin t}{t}$ has

A. a minimum **B.** a discontinuity

C. a point of inflection **D.** a maximum

Q2.10.2 MCQ 2010 1M

The value of the quantity, where

$$P = \int_0^1 xe^x \, dx$$

is

A. 0 **B.** 1 **C.** e **D.** 1/e

Q2.09.1 MCQ 2009 2M

If $f(x, y)$ is continuous function defined over $(x, y) \in [0, 1] \times [0, 1]$. Given two constraints, $x > y^2$ and $y > x^2$, the volume under $f(x, y)$ is

A. $\int_{y=0}^{y=1} \int_{x=y^2}^{x=\sqrt{y}} f(x, y) \, dx \, dy$

B. $\int_{y=x^2}^{y=1} \int_{x=y^2}^{x=1} f(x, y) \, dx \, dy$

C. $\int_{y=0}^{y=1} \int_{x=0}^{x=1} f(x, y) \, dx \, dy$

D. $\int_{x=0}^{y=\sqrt{x}} \int_{x=0}^{x=\sqrt{y}} f(x, y) \, dx \, dy$

Q2.07.1 MCQ 2007 1M

Consider the function $f(x) = (x^2 - 4)^2$ where x is a real number. Then the function has

A. Only one minimum **B.** Only two minima

C. Three minima **D.** Three maxima

Q2.07.2 MCQ 2007 2M

The integral $\frac{1}{2\pi} \int_0^{2\pi} \sin(t - \tau) \cos \tau \, d\tau$ equals

A. $\sin t \cos t$ **B.** 0

C. $\frac{1}{2} \cos t$ **D.** $\frac{1}{2} \sin t$

Q2.06.1 MCQ 2006 1M

The expression $V = \int_0^H \pi R^2 \left(1 - \frac{h}{H}\right)^2 dh$ for the volume of a cone is equal to _____.

A. $\int_0^R \pi R^2 \left(1 - \frac{h}{H}\right)^2 dr$ **B.** $\int_0^R \pi R^2 \left(1 - \frac{h}{H}\right)^2 dh$

C. $\int_0^H 2\pi rH \left(1 - \frac{r}{R}\right) dh$ **D.** $\int_0^R 2\pi rH \left(1 - \frac{r}{R}\right) dr$

Q2.05.1 MCQ 2005 2M

If $S = \int_1^\infty x^{-3} \, dx$ then S has the value

A. $\frac{-1}{3}$ **B.** $\frac{1}{4}$ **C.** $\frac{1}{2}$ **D.** 1

Q2.05.2 MCQ 2005 1M

For the function $f(x) = x^2 e^{-x}$, the maximum occurs when x is equal to

A. 2 **B.** 1 **C.** 0 **D.** -1

Instrumentation Engineering

Q2.26.1 MCQ 2026 1M

Which one among the following statements is true for the function $f(x) = x|x|$?

A. $f(x)$ is discontinuous but not differentiable at $x = 0$

B. $f(x)$ is discontinuous and differentiable at $x = 0$

C. $f(x)$ is continuous but not differentiable at $x = 0$

D. $f(x)$ is continuous and differentiable at $x = 0$

Q2.26.2 MSQ 2026 2M

Given $f(x, y) = x^2 - 2xy + y^2$
The complete contour of the equation $f(x, y) = 1$ is described by the option(s) _____.

- [ ] **A.** A line, $x - y = 1$
- [ ] **B.** A circle with radius 1
- [ ] **C.** An ellipse with length of major axis equal to 1
- [ ] **D.** A line, $x - y = -1$

Q2.25.1 MCQ 2025 2M

The value of the integral $\int_{-\pi}^{\pi} \left(\cos^6 x + \cos^4 x\right) dx$ is

A. $\frac{\pi}{2}$ **B.** $\frac{5\pi}{8}$ **C.** $\frac{11\pi}{8}$ **D.** $\frac{9\pi}{8}$

Q2.23.1 MCQ 2023 1M

What is $\lim_{x\to 0} f(x)$, where $f(x) = x \sin \frac{1}{x}$?

A. 0 **B.** 1 **C.** ∞ **D.** Limit does not exist

Q2.23.2 NAT 2023 2M

Consider the real-valued function $g(x) = \max\{(x-2)^2, -2x+7\}$, where $x \in (-\infty, \infty)$. The minimum value attained by $g(x)$ is _____ **(rounded off to one decimal place)**.

Q2.22.1 NAT 2022 1M

The global minimum of $x^3 e^{-|x|}$ for $x \in (-\infty, \infty)$ occurs at $x =$ _____ (round off to one decimal place)

Q2.21.1 NAT 2021 1M

Consider the function $f(x) = -x^2 + 10x + 100$. The minimum value of the function in the interval $[5, 10]$ is _____.

Q2.21.2 MCQ 2021 1M

Consider the sequence $x_n = 0.5x_{n-1} + 1, n = 1, 2, \ldots$ with $x_0 = 0$. Then $\lim_{n\to\infty} x_n$ is

A. 2 **B.** ∞ **C.** 1 **D.** 0

Q2.20.1 MCQ 2020 2M

A straight line drawn on an x-y plane intercepts the x-axis at -0.5 and the y-axis at 1. The equation that describes this line is _____.

A. $y = 2x + 1$ **B.** $y = x - 0.5$
C. $y = -0.5x + 1$ **D.** $y = 0.5x - 1$

Q2.20.2 NAT 2020 2M

Consider the function $f(x, y) = x^2 + y^2$. The minimum value the function attains on the line $x + y = 1$ (rounded off to one decimal place) is _____.

Q2.19.1 MCQ 2019 2M

The curve $y = f(x)$ is such that the tangent to the curve at every point (x, y) has a Y-axis intercept c, given by $c = -y$. then, $f(x)$ is proportional to

A. x^{-1} **B.** x^2 **C.** x^3 **D.** x^4

Q2.18.1 NAT 2018 1M

Consider two functions $f(x) = (x-2)^2$ and $g(x) = 2x - 1$, where x is real. The smaller value of x for which $f(x) = g(x)$ is _____.

Q2.18.2 MCQ 2018 2M

Consider the following equations

$$\frac{\partial V(x, y)}{\partial x} = px^2 + y^2 + 2xy$$

$$\frac{\partial V(x, y)}{\partial y} = x^2 + qy^2 + 2xy$$

where p and q are constants, $V(x, y)$ that satisfies the above equations is

A. $p\frac{x^3}{3} + q\frac{y^3}{3} + 2xy + 6$

B. $p\frac{x^3}{3} + q\frac{y^3}{3} + 5$

C. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2 + xy$

D. $p\frac{x^3}{3} + q\frac{y^3}{3} + x^2y + xy^2$

Q2.18.3 MCQ 2018 1M

For $0 \le x \le 2\pi$, $\sin x$ and $\cos x$ are both decreasing functions in the interval _____.

A. $\left(0, \frac{\pi}{2}\right)$ **B.** $\left(\frac{\pi}{2}, \pi\right)$

C. $\left(\pi, \frac{3\pi}{2}\right)$ **D.** $\left(\frac{3\pi}{2}, 2\pi\right)$

Q2.16.1 NAT 2016 1M

A straight line of the form $y = mx + c$ passes through the origin and the point $(x, y) = (2, 6)$. The value of m is _____.

Q2.16.2 NAT 2016 1M

$\lim_{n\to\infty}\left(\sqrt{n^2+n} - \sqrt{n^2+1}\right)$ is _____.

Q2.16.3 NAT 2016 2M

Let $f: [-1, 1] \to \mathbb{R}$, where $f(x) = 2x^3 - x^4 - 10$. The minimum value of $f(x)$ is _____.

Q2.15.1 MCQ 2015 1M

The double integral $\int_0^a \int_0^y f(x, y)\, dx\, dy$ is equivalent to

A. $\int_0^x \int_0^y f(x, y)\, dx\, dy$

B. $\int_0^a \int_x^y f(x, y)\, dx\, dy$

C. $\int_0^a \int_x^a f(x, y)\, dy\, dx$

D. $\int_0^a \int_0^a f(x, y)\, dx\, dy$

Q2.14.1 MCQ 2014 1M

Given

$$x(t) = 3\sin(1000\pi t) \quad \text{and}$$
$$y(t) = 5\cos\left(1000\pi t + \frac{\pi}{4}\right)$$

The x-yplot will be

A. a circle

B. a mulit-loop closed curve

C. a hyperbola

D. an ellipse

Q2.13.1 MCQ 2013 1M

The type of the partial differential equation $\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}$ is

A. Parabolic **B.** Elliptic

C. Hyperbolic **D.** Nonlinear

Q2.11.1 MCQ 2011 2M

The series $\sum_{m=0}^{\infty} \frac{1}{4^m}(x-1)^{2m}$ converges for

A. $-2 < x < 2$ **B.** $-1 < x < 3$

C. $-3 < x < 1$ **D.** $x < 3$

Q2.08.1 MCQ 2008 2M

The expression $e^{-\ln x}$ for $x > 0$ is equal to

A. $-x$ **B.** x **C.** x^{-1} **D.** $-x^{-1}$

Q2.08.2 MCQ 2008 2M

Consider the function $y = x^2 - 6x + 9$. The maximum value of y obtained when x varies over the interval 2 to 5 is

A. 1 **B.** 3 **C.** 4 **D.** 9

Q2.08.3 MCQ 2008 1M

Given $y = x^2 + 2x + 10$ the value of $\left.\frac{dy}{dx}\right|_{x=1}$ is equal to

A. 0 **B.** 4

C. 12 **D.** 13

Q2.08.4 MCQ 2008 1M

$\lim_{x\to 0} \frac{\sin x}{x}$ is

A. indeterminate **B.** 0

C. 1 **D.** ∞

Q2.07.1 MCQ 2007 2M

The value of $\int_0^{\infty} \int_0^{\infty} e^{-x^2} e^{-y^2}\, dx\, dy$ is

A. $\frac{\sqrt{\pi}}{2}$ **B.** $\sqrt{\pi}$ **C.** π **D.** $\frac{\pi}{4}$

Q2.07.2 MCQ 2007 2M

For real x, the maximum value of $\frac{e^{\sin x}}{e^{\cos x}}$ is

A. 1 **B.** e **C.** $e^{\sqrt{2}}$ **D.** ∞

Q2.07.3 MCQ 2007 2M

Consider the function $f(x) = |x|^3$, where x is real. Then the function $f(x)$ at $x = 0$ is

A. continuous but not differentiable

B. once differentiable but not twice.

C. twice differentiable but not thrice.

D. thrice differentiable

Q2.05.1 MCQ 2005 1M

$f = a_0x^n + a_1x^{n-1}y + \cdots + a_{n-1}xy^{n-1} + a_ny^n$ where a_i ($i = 0$ to n) are constants then $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}$ is

A. $\frac{f}{n}$ **B.** $\frac{n}{f}$

C. nf **D.** $n\sqrt{f}$

Q2.05.2 MCQ 2005 1M

The value of the integral $\int_{-1}^{1} \frac{1}{x^2}\,dx$ is

A. 2 **B.** does not exists

C. -2 **D.** ∞

Q2.01.1 MCQ 2001 1M

$\lim_{x\to\frac{\pi}{4}} \frac{\sin 2\left(x-\frac{\pi}{4}\right)}{x-\frac{\pi}{4}} =$ _____

A. 0 **B.** $\frac{1}{2}$ **C.** 1 **D.** 2

Q2.99.1 MCQ 1999 1M

$\lim_{x\to 0} \frac{1}{10}\frac{1-e^{-j5x}}{1-e^{-jx}} =$ _____

A. 0 **B.** 1.1 **C.** 0.5 **D.** 1

Computer Science & IT

Q2.26.1 MSQ 2026 2M

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as follows:

$$f(x) = \left(\frac{|x|}{2} - x\right)\left(x - \frac{|x|}{2}\right)$$

Which of the following statements is/are true?

- **A.** f has a local maximum
- **B.** f has a local minimum
- **C.** f' is continuous over $\mathbb{R}$
- **D.** f' is not differentiable over $\mathbb{R}$

Q2.26.2 MSQ 2026 1M

For a real number a, let $I(a) = \int_{-1}^{1}(3x^2 - ax + 1)\,dx$. Which of the following statements is/are true?

- **A.** The value of $I(a)$ is independent of the value of a
- **B.** The value of $I(a)$ can vary with the value of a
- **C.** There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a positive real number
- **D.** There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a negative real number

Q2.25.1 MCQ 2025 1M

The value of x such that $x > 1$, satisfying the equation $\int_1^x t \ln t\,dt = \frac{1}{4}$ is

A. $\sqrt{e}$ **B.** e **C.** e^2 **D.** $e - 1$

Q2.24.1 MCQ 2024 1M

Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

A. $\{-1, 1, 2\}$ **B.** $\{-2, -1, 1\}$ **C.** $\{0, 1\}$ **D.** $\{-1, 0, 1\}$

Q2.24.2 MCQ 2024 1M

Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2 - x)$$

Which one of the following options is the CORRECT value of $\int_0^2 f(x)\,dx$?

A. 0 **B.** 1 **C.** 2 **D.** -1

Q2.23.1 MSQ 2023 1M

Let

$$f(x) = x^3 + 15x^2 - 33x - 36$$

be a real-valued function.

Which of the following statements is/are TRUE?

- **A.** $f(x)$ does not have a local maximum.
- **B.** $f(x)$ has a local maximum.
- **C.** $f(x)$ does not have a local minimum.
- **D.** $f(x)$ has a local minimum.

Q2.23.2 NAT 2023 1M

The value of the definite integral

$$\int_{-3}^{3}\int_{-2}^{2}\int_{-1}^{1}(4x^2y - z^3)\,dz\,dy\,dx$$

is _____. (Rounded off to the nearest integer)

Q2.22.1 NAT 2022 1M

The value of the following limit is _____.

$$\lim_{x\to 0^+}\frac{\sqrt{x}}{1-e^{2\sqrt{x}}}$$

Q2.21.1 NAT 2021 1M

Consider the following expression.

$$\lim_{x\to -3}\frac{\sqrt{2x+22}-4}{x+3}$$

The value of the above expression (rounded to 2 decimal places) is _____.

Q2.21.2 NAT 2021 1M

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous function on the interval $[-3, 3]$ and a differentiable function in the interval $(-3, 3)$ such that for every x in the interval, $f'(x) \le 2$. If $f(-3) = 7$, then $f(3)$ is at most _____.

Q2.20.1 MCQ 2020 1M

Consider the functions

I. e^{-x}

II. $x^2 - \sin x$

III. $\sqrt{x^3+1}$

Which of the above functions is/are increasing everywhere in $[0, 1]$?

A. III only

B. II only

C. II and III only

D. I and III only

Q2.19.1 MCQ 2019 1M

Compute $\lim_{x\to 3}\frac{x^4-81}{2x^2-5x-3}$

A. 1 **B.** 53/12

C. 108/7 **D.** Limit does not exist

Q2.18.1 MCQ 2018 1M

Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0, 1, 2, \ldots$?

A. $\frac{3}{(1-x)^2}$ **B.** $\frac{3x}{(1-x)^2}$ **C.** $\frac{2-x}{(1-x)^2}$ **D.** $\frac{3-x}{(1-x)^2}$

Q2.18.2 NAT 2018 1M

The value of $\int_0^{\pi/4} x\cos(x^2)\,dx$ correct to three decimal places (assuming that $\pi = 3.14$) is _____.

Q2.17.1 MCQ 2017 2M

The value of $\lim_{x\to 1}\frac{x^7-2x^5+1}{x^3-3x^2+2}$

A. is 0 **B.** is -1

C. is 1 **D.** does not exist

Q2.17.2 MCQ 2017 2M

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to

A. the maximum of x and y

B. the minimum of x and y

C. 1

D. none of the above

Q2.17.3 MCQ 2017 1M

If $f(x) = R\sin\left(\frac{\pi x}{2}\right) + S$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x)dx = \frac{2R}{\pi}$, then the constants R and S are, respectively

A. $\frac{2}{\pi}$ and $\frac{16}{\pi}$ **B.** $\frac{2}{\pi}$ and 0

C. $\frac{4}{\pi}$ and 0 **D.** $\frac{4}{\pi}$ and $\frac{16}{\pi}$

Q2.17.4 NAT 2017 1M

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b. The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b =$ _____.

Q2.17.5 MCQ 2017 2M

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

A. 0 **B.** 1 **C.** 2 **D.** 3

Q2.16.1 NAT 2016 1M

$\lim_{x\to 4} \dfrac{\sin(x-4)}{x-4} =$ _____.

Q2.16.2 NAT 2016 1M

Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is _____.

Q2.15.1 MCQ 2015 1M

If $g(x) = 1 - x$ and $h(x) = \dfrac{x}{x-1}$, then $\dfrac{g(h(x))}{h(g(x))}$ is:

A. $\dfrac{h(x)}{g(x)}$ **B.** $\dfrac{-1}{x}$ **C.** $\dfrac{g(x)}{h(x)}$ **D.** $\dfrac{x}{(1-x)^2}$

Q2.15.2 MCQ 2015 1M

$\lim_{x\to\infty} x^{1/x}$ is

A. ∞ **B.** 0 **C.** 1 **D.** Not defined

Q2.15.3 NAT 2015 2M

$\sum_{x=1}^{99} \dfrac{1}{x(x+1)} =$ _____.

Q2.15.4 NAT 2015 2M

$\int_{1/\pi}^{2/\pi} \dfrac{\cos(1/x)}{x^2} dx =$ _____.

Q2.15.5 MCQ 2015 1M

The value of $\lim_{x\to\infty}(1+x^2)^{e^{-x}}$ is

A. 0 **B.** $\dfrac{1}{2}$ **C.** 1 **D.** ∞

Q2.14.1 NAT 2014 1M

If $\int_0^{2\pi} |x \sin x| dx = k\pi$, then the value of k is equal to _____.

Q2.14.2 MCQ 2014 2M

The value of the integral given below is

$$\int_0^{\pi} x^2 \cos x dx$$

A. -2π **B.** π **C.** $-\pi$ **D.** 2π

Q2.14.3 MCQ 2014 1M

A non-zero polynomial $f(x)$ of degree 3 has roots at $x = 1, x = 2$ and $x = 3$. Which one of the following must be TRUE?

A. $f(0)f(4) < 0$ **B.** $f(0)f(4) > 0$

C. $f(0) + f(4) > 0$ **D.** $f(0) + f(4) < 0$

Q2.13.1 MCQ 2013 1M

Which one of the following functions is continuous at $x = 3$?

A. $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x-1, & \text{if } x > 3 \\ \dfrac{x+3}{3}, & \text{if } x < 3 \end{cases}$

B. $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8-x, & \text{if } x \neq 3 \end{cases}$

C. $f(x) = \begin{cases} x+3, & \text{if } x \leq 3 \\ x-4, & \text{if } x > 3 \end{cases}$

D. $f(x) = \dfrac{1}{x^3 - 27}$, if $x \neq 3$

Q2.12.1 MCQ 2012 1M

Consider the function $f(x) = \sin(x)$ in the interval $x \in [\pi/4, 7\pi/4]$. The number and location(s) of the local minima of this function are

A. One, at $\pi/2$

B. One, at $3\pi/2$

C. Two, at $\pi/2$ and $3\pi/2$

D. Two, at $\pi/4$ and $3\pi/2$

Q2.11.1 MCQ 2011 2M

Given $i = \sqrt{-1}$, what will be the evaluation of the definite integral $\int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$?

A. 0 **B.** 2 **C.** $-i$ **D.** i

Q2.10.1 MCQ 2010 1M

What is the value of $\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^{2n}$?

A. 0 **B.** e^{-2} **C.** $e^{-1/2}$ **D.** 1

Q2.09.1 MCQ 2009 1M

$\int_0^{\pi/4} (1 - \tan x)/(1 + \tan x) dx$ evaluates to

A. 0 **B.** 1 **C.** $\ln 2$ **D.** $\frac{1}{2} \ln 2$

Q2.05.1 MCQ 2005 1M

$\lim_{x\to 0} \frac{\sin^2 x}{x}$ is

A. 0 **B.** ∞ **C.** 1 **D.** Not defined

Q2.05.2 MCQ 2005 2M

The value of the integral $\int_0^{\pi} \theta \sin^3 \theta d\theta$ is

A. $\frac{2\pi}{3}$ **B.** $\frac{\pi}{3}$ **C.** $\frac{\pi}{6}$ **D.** $\frac{\pi}{2}$

Q2.04.1 MCQ 2004 1M

The following integral $\int_1^e x \ln x dx$ is equal to

A. $\frac{e^2 + 1}{4}$ **B.** $\frac{e^2 - 1}{4}$ **C.** $\frac{e^2 + 1}{2}$ **D.** $\frac{e^2 - 1}{2}$

Q2.03.1 MCQ 2003 1M

The value of $\lim_{x\to 0} \left(\frac{-\sin x}{x}\right)$ is

A. 1 **B.** -1 **C.** 0 **D.** ∞

Mechanical Engineering

Q2.26.1 MCQ 2026 1M

Domain A is bounded by curve $x^2 = 4y$, ordinate $x = 2$ and x axis. The value of $\iint_A y dx dy$ is

A. $\frac{1}{5}$ **B.** $\frac{1}{3}$

C. $\frac{5}{12}$ **D.** $\frac{1}{2}$

Q2.25.1 MCQ 2025 2M

In the closed interval $[0, 3]$, the minimum value of the function f given below is

$$f(x) = 2x^3 - 9x^2 + 12x$$

A. 0 **B.** 4 **C.** 5 **D.** 9

Q2.23.1 MCQ 2023 1M

The figure shows the plot of a function over the interval $[-4, 4]$. Which one of the options given CORRECTLY identifies the function?

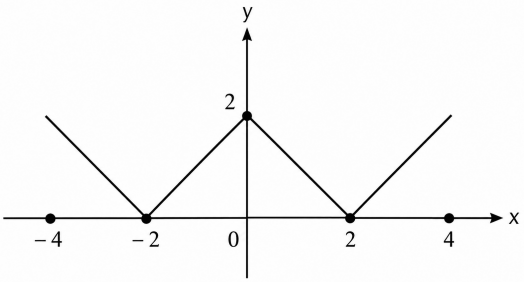


Fig. Q2.23.1

A. |2 - x| **B.** |2 - |x||

C. |2 + |x|| **D.** 2 - |x|

Q2.23.2 NAT 2023 2M

The smallest perimeter that a rectangle with area of 4 square units can have is _____ units. (Answer in integer)

Q2.22.1 MCQ 2022 1M

The limit $p = \lim_{x\to\pi}\left(\frac{x^2+\alpha x+2\pi^2}{x-\pi+2\sin x}\right)$ has a finite value for a real α. The value of α and the corresponding limit p are

A. $\alpha = -3\pi$, and $p = \pi$

B. $\alpha = -2\pi$, and $p = 2\pi$

C. $\alpha = \pi$, and $p = \pi$

D. $\alpha = 2\pi$, and $p = 3\pi$

Q2.22.2 MCQ 2022 1M

Given $\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$. If a,b are positive integers, the value of $\int_{-\infty}^{\infty} e^{-a(x+b)^2}dx$ is _____.

A. $\sqrt{\pi a}$ **B.** $\sqrt{\frac{\pi}{a}}$

C. $b\sqrt{\pi a}$ **D.** $b\sqrt{\frac{\pi}{a}}$

Q2.21.1 MCQ 2021 2M

The value of $\int_0^{\pi/2}\int_0^{\cos\theta} r\sin\theta\, dr\, d\theta$ is

A. π **B.** $\frac{1}{6}$ **C.** 0 **D.** $\frac{4}{3}$

Q2.21.2 MCQ 2021 1M

The value of $\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)$ is

A. 1 **B.** $\frac{1}{3}$ **C.** $\frac{1}{4}$ **D.** $\frac{1}{2}$

Q2.21.3 MCQ 2021 2M

Let $f(x) = x^2 - 2x + 2$ be a continuous function defined on $x \in [1, 3]$. The point x at which the tangent of $f(x)$ becomes parallel to the straight line joining $f(1)$ and $f(3)$ is

A. 3 **B.** 1 **C.** 0 **D.** 2

Q2.20.1 MCQ 2020 1M

Let $I = \int_{x=0}^{1}\int_{y=0}^{x^2} xy^2\, dydx$ .Then here I may also be expressed as

A. $\int_{y=0}^{1}\int_{x=0}^{\sqrt{y}} yx^2\, dxdy$ **B.** $\int_{y=0}^{1}\int_{x=\sqrt{y}}^{1} yx^2\, dxdy$

C. $\int_{y=0}^{1}\int_{x=0}^{\sqrt{y}} xy^2\, dxdy$ **D.** $\int_{y=0}^{1}\int_{x=\sqrt{y}}^{1} xy^2\, dxdy$

Q2.20.2 MCQ 2020 1M

The value of $\lim_{x\to 1}\left(\frac{1-e^{-c(1-x)}}{1-xe^{-c(1-x)}}\right)$ is

A. c **B.** $\frac{c+1}{c}$

C. $c+1$ **D.** $\frac{c}{c+1}$

Q2.19.1 MCQ 2019 1M

A parabola $x = y^2$ with $0 \le x \le 1$ is shown in the figure. The volume of the solid of rotation obtained by rotating the shaded area by 360° around the x-axis is

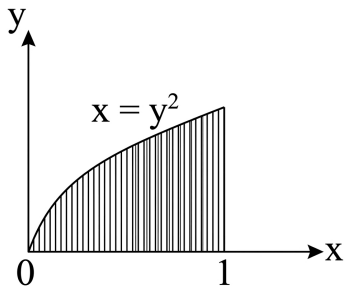


Fig. Q2.19.1

A. π **B.** $\frac{\pi}{4}$

C. 2π **D.** $\frac{\pi}{2}$

Q2.19.2 NAT 2019 2M

The value of the following definite integral $\int_1^e (x\ln x)dx$ is _____. (round off to three decimal places)

Q2.19.3 MCQ 2019 2M

The derivative of $f(x) = \cos(x)$ can be estimated using the approximation

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

The percentage error is calculated as

$$\left(\frac{\text{Exact value} - \text{approximate value}}{\text{Exact value}}\right) \times 100.$$

The percentage error in the derivative of $f(x)$ at $x = \pi/6$ radian, choosing $h = 0.1$ radian, is

A. $> 5\%$

B. $> 0.1\%$ and $< 1\%$

C. $> 1\%$ and $< 5\%$

D. $< 0.1\%$

Q2.18.1 MCQ 2018 1M

According to Mean Value Theorem, for a continuous function $f(x)$ in the interval $[a, b]$, there exists a value ξ in this interval such that $\int_a^b f(x)dx =$

A. $f(\xi)(b-a)$ **B.** $f(b)(\xi-a)$

C. $f(a)(b-\xi)$ **D.** 0

Q2.17.1 MCQ 2017 1M

The value of $\lim_{x\to 0}\left(\frac{x^3-\sin(x)}{x}\right)$ is

A. 0 **B.** 3 **C.** 1 **D.** -1

Q2.17.2 MCQ 2017 2M

A parametric curve defined by $x = \cos\left(\frac{\pi u}{2}\right)$ and $y = \sin\left(\frac{\pi u}{2}\right)$ in the range $0 \le u \le 1$ is rotated about the X-axis by 360 degrees. Area of the surface generated is

A. $\frac{\pi}{2}$ **B.** π **C.** 2π **D.** 4π

Q2.16.1 NAT 2016 2M

Consider the function $f(x) = 2x^3 - 3x^2$ in the domain $[-1, 2]$. The global minimum of $f(x)$ is _____.

Q2.16.2 MCQ 2016 1M

The values of x for which the function $f(x) = \frac{x^2-3x-4}{x^2+3x-4}$ is NOT continuous are

A. 4 and -1 **B.** 4 and 1

C. -4 and 1 **D.** -4 and -1

Q2.16.3 MCQ 2016 1M

$\lim_{x\to 0}\frac{\log_e(1+4x)}{e^{3x}-1}$ is equal to

A. 0 **B.** $\frac{1}{12}$ **C.** $\frac{4}{3}$ **D.** 1

Q2.16.4 MCQ 2016 2M

$\lim_{x\to\infty}\left(\sqrt{x^2+x-1}-x\right)$ is

A. 0 **B.** ∞ **C.** $1/2$ **D.** $-\infty$

Q2.15.1 NAT 2015 1M

The value of $\lim_{x\to 0}\left(\frac{-\sin x}{2\sin x + x\cos x}\right)$ is _____.

Q2.15.2 MCQ 2015 1M

The value of $\lim_{x\to 0}\frac{1-\cos(x^2)}{2x^4}$ is

A. 0 **B.** $\frac{1}{2}$ **C.** $\frac{1}{4}$ **D.** undefined

Q2.15.3 NAT 2015 2M

Consider a spatial curve in three-dimensional space given in parametric form by $x(t) = \cos t$, $y(t) = \sin t$, $z(t) = \frac{2}{\pi}t$, $0 \le t \le \frac{\pi}{2}$. The length of the curve is _____. (rounded off to four decimal places).

Q2.15.4 NAT 2015 1M

Consider an ant crawling along the curve $(x-2)^2 + y^2 = 4$, where x and y are in meters. The ant starts at the point $(4, 0)$ and moves counter-clockwise with a speed of 1.57 meters per second. The time taken by the ant to reach the point $(2, 2)$ is _____ (in seconds).

Q2.15.5 MCQ 2015 1M

At $x = 0$, the function $f(x) = |x|$ has

A. a minimum

B. a maximum

C. a point of inflexion

D. neither a maximum nor minimum

Q2.14.1 MCQ 2014 1M

$\lim_{x\to 0} \frac{x - \sin x}{1 - \cos x}$ is

A. 0 **B.** 1 **C.** 3 **D.** not defined

Q2.14.2 MCQ 2014 1M

$\lim_{x\to 0} \left(\frac{e^{2x} - 1}{\sin(4x)}\right)$ is equal to

A. 0 **B.** 0.5 **C.** 1 **D.** 2

Q2.14.3 MCQ 2014 1M

If a function is continuous at a point,

A. the limit of the function may not exist at the point

B. The function must be derivable at the point

C. the limit of the function at the point tends to infinity

D. the limit must exist at the point and the value of limit should be same as the value of the function at the point.

Q2.14.4 MCQ 2014 1M

The value of the integral $\int_0^2 \frac{(x-1)^2 \sin(x-1)}{(x-1)^2 + \cos(x-1)}\, dx$ is

A. 3 **B.** 0 **C.** -1 **D.** -2

Q2.14.5 MCQ 2014 2M

The value of the integral $\int_0^2 \int_0^x e^{x+y}\, dy\, dx$ is

A. $\frac{1}{2}(e-1)$ **B.** $\frac{1}{2}(e^2-1)^2$

C. $\frac{1}{2}(e^2-e)$ **D.** $\frac{1}{2}\left(e - \frac{1}{e}\right)^2$

Q2.13.1 MCQ 2013 2M

The value of the definite integral $\int_1^e \sqrt{x}\ln(x)dx$ is

A. $\frac{4}{9}\sqrt{e^3} + \frac{2}{9}$ **B.** $\frac{2}{9}\sqrt{e^3} - \frac{4}{9}$

C. $\frac{2}{9}\sqrt{e^3} + \frac{4}{9}$ **D.** $\frac{4}{9}\sqrt{e^3} - \frac{2}{9}$

Q2.12.1 MCQ 2012 1M

At $x = 0$, the function $f(x) = x^3 + 1$ has

A. a maximum value

B. a minimum value

C. a singularity

D. a point of inflection

Q2.12.2 MCQ 2012 2M

A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

A. 8 meters **B.** 10 meters

C. 12 meters **D.** 14 meters

Q2.11.1 MCQ 2011 1M

If $f(x)$ is even function and a is a positive real number, then $\int_{-a}^{a} f(x)dx$ equals _____.

A. 0 **B.** a

C. $2a$ **D.** $2\int_0^a f(x)dx$

Q2.10.1 MCQ 2010 1M

The infinite series $f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + - \ldots$ Converges to

A. $\cos(x)$ **B.** $\sin(x)$

C. $\sin\ h(x)$ **D.** e^x

Q2.10.2 MCQ 2010 2M

The value of the integral $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

A. $-\pi$ **B.** $-\frac{\pi}{2}$ **C.** $\frac{\pi}{2}$ **D.** π

Q2.10.3 MCQ 2010 2M

The function $y = |2 - 3x|$

A. is continuous $\forall\, x \in R$ and differentiable $\forall\, x \in R$

B. is continuous $\forall\, x \in R$ and differentiable $\forall\, x \in R$ except at $x = \frac{3}{2}$

C. is continuous $\forall\, x \in R$ and differentiable $\forall\, x \in R$ except at $x = \frac{2}{3}$

D. is continuous $\forall\, x \in R$ and except at $x = 3$ and differentiable $\forall\, x \in R$

Q2.10.4 MCQ 2010 1M

The parabolic arc $y = \sqrt{x}, 1 \le x \le 2$ is revolved around the x-axis. The volume of the solid of revolution is

A. $\frac{\pi}{4}$ **B.** $\frac{\pi}{2}$ **C.** $\frac{3\pi}{4}$ **D.** $\frac{3\pi}{2}$

Q2.09.1 MCQ 2009 2M

The distance between the origin and the point nearest to it on the surface $z^2 = 1 + xy$ is

A. 1 **B.** $\frac{\sqrt{3}}{2}$ **C.** $\sqrt{3}$ **D.** 2

Q2.08.1 MCQ 2008 2M

Consider the shaded triangular region P shown in the figure. What is $\iint_P xy\, dx\, dy$?

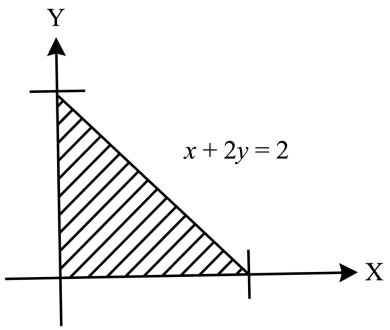


Fig. Q2.08.1

A. $\frac{1}{6}$ **B.** $\frac{2}{9}$ **C.** $\frac{7}{16}$ **D.** 1

Q2.08.2 MCQ 2008 1M

In the Taylor series expansion of e^x about $x = 2$, the coefficient of $(x-2)^4$ is

A. $\frac{1}{4!}$ **B.** $\frac{2^4}{4!}$ **C.** $\frac{e^2}{4!}$ **D.** $\frac{e^4}{4!}$

Q2.08.3 MCQ 2008 1M

The value of $\lim_{x\to 8} \frac{x^{1/3} - 2}{x - 8}$ is

A. $\frac{1}{16}$ **B.** $\frac{1}{12}$ **C.** $\frac{1}{8}$ **D.** $\frac{1}{4}$

Q2.08.4 MCQ 2008 2M

Which of the following integrals is unbounded?

A. $\int_0^{\pi/4} \tan x\, dx$ **B.** $\int_0^{\infty} \frac{1}{1+x^2}\, dx$

C. $\int_0^{\infty} x \cdot e^{-x}\, dx$ **D.** $\int_0^{1} \frac{1}{1-x}\, dx$

Q2.08.5 MCQ 2008 1M

The length of the curve $y = \frac{2}{3}x^{3/2}$ between $x = 0$ & $x = 1$ is

A. 0.27 **B.** 0.67 **C.** 1 **D.** 1.22

Q2.08.6 MCQ 2008 2M

Let $f = y^x$. What is $\frac{\partial^2 f}{\partial x\, \partial y}$ at $x = 2, y = 1$?

A. 0 **B.** $\ln 2$ **C.** 1 **D.** $\frac{1}{\ln 2}$

Q2.07.1 MCQ 2007 1M

$\lim_{x\to 0} \frac{e^x - \left(1 + x + \frac{x^2}{2}\right)}{x^3} =$ _____.

A. 0 **B.** $\frac{1}{6}$ **C.** $\frac{1}{3}$ **D.** 1

Q2.07.2 MCQ 2007 2M

If $y = x + \sqrt{x + \sqrt{x + \sqrt{x + \ldots \infty}}}$, then $y(2) =$ _____.

A. 4 (or) 1 **B.** 4 only

C. 1 only **D.** Undefined

Civil Engineering

Q2.26.1 MCQ 2026 2M

Let $f(x)$ be a continuous function defined in $[0, 2] \to \mathbb{R}$ and satisfying the equation $\int_0^2 f(x)[x - f(x)]dx = \frac{2}{3}$. The value of $f(1)$ is

A. 1 **B.** 2 **C.** $\frac{1}{2}$ **D.** 0

Q2.25.1 MCQ 2025 1M

Integration of $\ln(x)$ with x i.e $\int \ln(x)\,dx =$ _____.

A. $x \cdot \ln(x) - x +$ Constant **B.** $x - \ln(x) +$ Constant

C. $x \cdot \ln(x) + x +$ Constant **D.** $\ln(x) - x +$ Constant

Q2.25.2 MCQ 2025 2M

The value of $\lim_{x\to\infty}(x - \sqrt{x^2 + x})$ is equal to

A. -1 **B.** -0.5 **C.** -2 **D.** 0

Q2.25.3 MSQ 2025 2M

Consider the function given below and pick one or more **CORRECT** statement(s) from the following choices. $f(x) = x^3 - \frac{15}{2}x^2 + 18x + 20$

- ☐ **A.** $f(x)$ has a local minimum at $x = 3$.
- ☐ **B.** $f(x)$ has a local maximum at $x = 3$.
- ☐ **C.** $f(x)$ has a local minimum at $x = 2$.
- ☐ **D.** $f(x)$ has a local maximum at $x = 2$.

Q2.25.4 NAT 2025 2M

The maximum value of the function $h(x) = -x^3 + 2x^2$ in the interval $[-1, 1.5]$ is equal to _____ (rounded off to 1 decimal place).

Q2.24.1 MCQ 2024 1M

The smallest positive root of the equation

$$x^5 - 5x^4 - 10x^3 + 50x^2 + 9x - 45 = 0$$

lies in the range

A. $0 < x \le 2$ **B.** $2 < x \le 4$

C. $6 \le x \le 8$ **D.** $10 \le x \le 100$

Q2.24.2 MCQ 2024 1M

The function $f(x) = x^3 - 27x + 4,\ 1 \le x \le 6$ has

A. Maxima point **B.** Minima point

C. Saddle point **D.** Inflection point

Q2.24.3 NAT 2024 2M

The expression for computing the effective interest rate (i_{eff}) using continuous compounding for a nominal interest rate of 5% is

$$i_{eff} = \lim_{m\to\infty}\left(1 + \frac{0.05}{m}\right)^m - 1$$

The effective interest rate (in percentage) is _____

Q2.23.1 MCQ 2023 1M

For the integral $I = \int_{-1}^{1} \frac{1}{x^2}\,dx$ which of the following statements is TRUE?

A. $I = 0$ **B.** $I = 2$

C. $I = -2$ **D.** The integral does not converge

Q2.23.2 MCQ 2023 2M

For the function $f(x) = e^x|\sin x|;\ x \in \mathbb{R}$, which of the following statements is/are TRUE?

A. The function is continuous at all x

B. The function is differentiable at all x

C. The function is periodic

D. The function is bounded

Q2.22.1 MSQ 2022 2M

Let $\max\{a, b\}$ denote the maximum of two real numbers a and b. Which of the following statement(s) is/are TURE about the function $f(x) = \max\{3 - x, x - 1\}$?

- ☐ **A.** It is continuous on its domain
- ☐ **B.** It has a local minimum at $x = 2$
- ☐ **C.** It has a local maximum at $x = 2$
- ☐ **D.** It is differentiable on its domain

Q2.22.2 MCQ 2022 1M

$\int \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots\right) dx$ is equal to

A. $\frac{1}{1+x} +$ Constant **B.** $\frac{1}{1+x^2} +$ Constant

C. $-\frac{1}{1-x} +$ Constant **D.** $-\frac{1}{1-x^2} +$ Constant

Q2.22.3 MCQ 2022 2M

Consider the polynomial $f(x) = x^3 - 6x^2 + 11x - 6$ on the domain S given by $1 \le x \le 3$. The first and second derivatives are $f'(x) and f''(x)$. Consider the following statements.

I. The given polynomial is zero at the boundary points $x = 1$ and $x = 3$

II. There exists one local maxima of $f(x)$ within the domain S.

III. The second derivative $f''(x) > 0$ throughout the domain S

IV. There exists one local minima of $f(x)$ within the domain S.

The correct option is:

A. Only statements I, II and III are correct

B. Only statements I, II and IV are correct

C. Only statements I and IV are correct

D. Only statements II and IV are correct

Q2.21.1 NAT 2021 1M

The volume determined from $\iiint_V 8xyz\,dV$ for $V = [2,3] \times [1,2] \times [0,1]$ will be (in integer) _____.

Q2.21.2 NAT 2021 1M

Consider the limit:

$$\lim_{x\to 1}\left(\frac{1}{\ln x} - \frac{1}{x-1}\right)$$

The limit (correct up to one decimal place) is _____.

Q2.21.3 NAT 2021 2M

The value (round off to one decimal place) of $\int_{-1}^{1} xe^{|x|}\,dx$ is _____.

Q2.21.4 MCQ 2021 1M

The value of $\lim_{x\to\infty} \frac{x\ln(x)}{1+x^2}$ is

A. ∞ **B.** 0.5 **C.** 0 **D.** 1.0

Q2.20.1 MCQ 2020 1M

The value of $\lim_{x\to\infty} \frac{x^2-5x+4}{4x^2+2x}$ is

A. 1 **B.** $\frac{1}{2}$ **C.** 0 **D.** $\frac{1}{4}$

Q2.20.2 MCQ 2020 1M

The value of $\lim_{x\to\infty} \frac{\sqrt{9x^2+2020}}{x+7}$ is

A. indeterminable **B.** 1 **C.** $\frac{7}{9}$ **D.** 3

Q2.20.3 MCQ 2020 1M

The area of an ellipse represented by an equation $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ is

A. $\frac{\pi ab}{4}$ **B.** $\frac{4\pi ab}{3}$

C. $\frac{\pi ab}{2}$ **D.** πab

Q2.19.1 MCQ 2019 1M

Which one of the following is correct?

A. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = 2$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = 1$

B. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = \infty$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = 1$

C. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = 1$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = 1$

D. $\lim_{x\to 0}\left(\frac{\sin 4x}{\sin 2x}\right) = 2$ and $\lim_{x\to 0}\left(\frac{\tan x}{x}\right) = \infty$

Q2.19.2 MCQ 2019 1M

For a small value of h, the Taylor series expansion for $f(x+h)$ is

A. $f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \ldots\infty$

B. $f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{3}f'''(x) + \ldots\infty$

C. $f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{3}f'''(x) + \ldots\infty$

D. $f(x) - hf'(x) + \frac{h^2}{2!}f''(x) - \frac{h^3}{3!}f'''(x) + \ldots\infty$

Q2.19.3 MCQ 2019 2M

Consider two functions: $x = \psi\ln\phi$ and $y = \phi\ln\psi$. Which one of the following is the correct expression for $\frac{\partial\psi}{\partial x}$?

A. $\frac{x\ln\phi}{\ln\phi\ln\psi - 1}$ **B.** $\frac{x\ln\psi}{\ln\phi\ln\psi - 1}$

C. $\frac{\ln\phi}{\ln\phi\ln\psi - 1}$ **D.** $\frac{\ln\psi}{\ln\phi\ln\psi - 1}$

Q2.19.4 MCQ 2019 2M

Which one of the following is NOT a correct statement?

A. The function $\sqrt[x]{x}$, $(x > 0)$, has the global minimum at $x = e$

B. The function $|x|$ has the global minima at $x = 0$

C. The function x^3 has neither global minima nor global maxima

D. The function $\sqrt[x]{x}$, $(x > 0)$, has the global maxima at $x = e$

Q2.19.5 MCQ 2019 1M

The following inequality is true for all x close to 0.

$$\left(2 - \frac{x^2}{3}\right) < \frac{x \sin x}{1 - \cos x} < 2$$

What is the value $\lim_{x \to 0} \frac{x \sin x}{1 - \cos x}$?

A. 2 **B.** 1 **C.** 0 **D.** $\frac{1}{2}$

Q2.18.1 MCQ 2018 1M

At the point $x = 0$, the function $f(x) = x^3$ has

A. local maximum

B. local minimum

C. both local maximum and minimum

D. neither local maximum nor local minimum

Q2.18.2 MCQ 2018 1M

The value of the integral $\int_0^{\pi} x \cos^2 x \, dx$ is

A. $\frac{\pi^2}{8}$ **B.** $\frac{\pi^2}{4}$ **C.** $\frac{\pi^2}{2}$ **D.** π^2

Q2.17.1 MCQ 2017 1M

The tangent to the curve represented by $y = x \ln x$ is required to have 45° inclination with the x-axis. The coordinates of the tangent point would be

A. $(1, 0)$ **B.** $(0, 1)$

C. $(1, 1)$ **D.** $(\sqrt{2}, \sqrt{2})$

Q2.17.2 MCQ 2017 2M

Consider the following definite integral

$$I = \int_0^1 \frac{(\sin^{-1} x)^2}{\sqrt{1 - x^2}} \, dx$$

The value of the integral is

A. $\frac{\pi^3}{24}$ **B.** $\frac{\pi^3}{12}$ **C.** $\frac{\pi^3}{48}$ **D.** $\frac{\pi^3}{64}$

Q2.17.3 NAT 2017 1M

$\lim_{x \to 0} \left(\frac{\tan x}{x^2 - x}\right)$ is equal to _____.

Q2.17.4 MCQ 2017 1M

Let x be a continuous variable defined over the interval $(-\infty, \infty)$ and $f(x) = e^{-x - e^{-x}}$. The integral $g(x) = \int f(x) \, dx$ is equal to

A. $e^{e^{-x}}$ **B.** $e^{-e^{-x}}$ **C.** $e^{-e^{x}}$ **D.** e^{-x}

Q2.17.5 MCQ 2017 1M

Let $W = f(x, y)$, where x and y are functions of t. Then, according to the chain rule, $\frac{dw}{dt}$ is equal to

A. $\frac{dw}{dx}\frac{dx}{dt} + \frac{dw}{dy}\frac{dt}{dt}$ **B.** $\frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t}$

C. $\frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt}$ **D.** $\frac{dw}{dx}\frac{\partial x}{\partial t} + \frac{dw}{dy}\frac{\partial y}{\partial t}$

Q2.16.1 MCQ 2016 2M

The quadratic approximation of $f(x) = x^3 - 3x^2 - 5$ at the point $x = 0$ is

A. $3x^2 - 6x - 5$ **B.** $-3x^2 - 5$

C. $-3x^2 + 6x - 5$ **D.** $3x^2 - 5$

Q2.16.2 MCQ 2016 2M

The value of $\int_0^{\infty} \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{\sin x}{x} dx$ is

A. $\frac{\pi}{2}$ **B.** π **C.** $\frac{3\pi}{2}$ **D.** 1

Q2.16.3 MCQ 2016 2M

The area of the region bounded by the parabola $y = x^2 + 1$ and the straight line $x + y = 3$ is

A. $\frac{59}{6}$ **B.** $\frac{9}{2}$ **C.** $\frac{10}{3}$ **D.** $\frac{7}{6}$

Q2.16.4 MCQ 2016 1M

The optimum value of the function $f(x) = x^2 - 4x + 2$ is

A. 2 (maximum) **B.** 2 (minimum)

C. −2 (maximum) **D.** −2 (minimum)

Q2.16.5 MCQ 2016 1M

What is the value of $\lim_{\substack{x\to 0\\ y\to 0}} \frac{xy}{x^2+y^2}$?

A. 1 **B.** -1 **C.** 0 **D.** Limit does not exist

Q2.16.6 MCQ 2016 2M

The angle of intersection of the curves $x^2 = 4y$ and $y^2 = 4x$ at point $(0, 0)$ is

A. $0°$ **B.** $30°$ **C.** $45°$ **D.** $90°$

Q2.16.7 NAT 2016 2M

The area between the parabola $x^2 = 8y$ and the straight line $y = 8$ is _____.

Q2.15.1 MCQ 2015 1M

While minimizing the function $f(x)$, necessary and sufficient conditions for a point x_0 to be a minima are:

A. $f'(x_0) > 0$ and $f''(x_0) = 0$

B. $f'(x_0) < 0$ and $f''(x_0) = 0$

C. $f'(x_0) = 0$ and $f''(x_0) < 0$

D. $f'(x_0) = 0$ and $f''(x_0) > 0$

Q2.15.2 MCQ 2015 1M

Given, $i = \sqrt{-1}$, the value of the definite integral, $I = \int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$ is:

A. 1 **B.** -1 **C.** i **D.** $-i$

Q2.15.3 MCQ 2015 1M

$\lim_{x\to\infty} \left(1 + \frac{1}{x}\right)^{2x}$ is equal to

A. e^{-2} **B.** e **C.** 1 **D.** e^2

Q2.14.1 MCQ 2014 1M

$\lim_{x\to\infty} \left(\frac{x + \sin x}{x}\right)$ equals

A. $-\infty$ **B.** 1 **C.** 0 **D.** ∞

Q2.14.2 MCQ 2014 1M

With reference to the conventional Cartesian (x, y) coordinate system, the vertices of a triangle have the following coordinates: $(x_1, y_1) = (1, 0)$; $(x_2, y_2) = (2, 2)$; and $(x_3, y_3) = (4, 3)$. The area of the triangle is equal to

A. $\frac{3}{2}$ **B.** $\frac{4}{5}$ **C.** $\frac{3}{4}$ **D.** $\frac{5}{2}$

Q2.14.3 MCQ 2014 2M

The expression $\lim_{a\to 0} \frac{x^a - 1}{a}$ is equal to

A. $\log x$ **B.** $x \log x$

C. 0 **D.** ∞

Q2.13.1 MCQ 2013 2M

The solution for $\int_0^{\pi/6} \cos^4 3\theta \sin^3 6\theta \, d\theta$ is:

A. 0 **B.** 1 **C.** $\frac{1}{15}$ **D.** $\frac{8}{3}$

Q2.12.1 MCQ 2012 1M

The infinite series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$ corresponds to

A. $\sec x$ **B.** $\cos x$

C. e^x **D.** $1 + \sin^2 x$

Q2.11.1 MCQ 2011 2M

What should be the value of λ such that the function defined below is continuous at $x = \frac{\pi}{2}$?

$$f(x) = \begin{cases} \frac{\lambda \cos x}{\frac{\pi}{2} - x}, & \text{if } x \neq \frac{\pi}{2} \\ 1, & \text{if } x = \frac{\pi}{2} \end{cases}$$

A. 0 **B.** 2π **C.** 1 **D.** $\frac{\pi}{2}$

Q2.11.2 MCQ 2011 2M

What is the value of the definite integral?

$$\int_0^a \frac{\sqrt{x}}{\sqrt{x} + \sqrt{a - x}} dx?$$

A. 0 **B.** $\frac{a}{2}$ **C.** a **D.** $2a$

Q2.10.1 MCQ 2010 1M

The $\lim_{x\to 0} \dfrac{\sin\left(\frac{2}{3}x\right)}{x}$ is

A. $\frac{2}{3}$ **B.** 1 **C.** $\frac{3}{2}$ **D.** ∞

Q2.10.2 MCQ 2010 2M

Given a function $f(x, y) = 4x^2 + 6y^2 - 8x - 4y + 8$, the optimal values of $f(x, y)$ is

A. a minimum equal to $\frac{10}{3}$

B. a maximum equal to $\frac{10}{3}$

C. a minimum equal to $\frac{8}{3}$

D. a maximum equal to $\frac{8}{3}$

Q2.10.3 MCQ 2010 2M

A parabolic cable is held between two supports at the same level. The horizontal span between the supports is L. The sag at the mid-span is h. The equation of the parabola is $y = 4h\dfrac{x^2}{L^2}$, where x is the horizontal coordinate and y is the vertical coordinate with the origin at the centre of the cable. The expression for the total length of the cable is

A. $\int_0^L \sqrt{1 + 64\dfrac{h^2x^2}{L^4}}dx$ **B.** $2\int_0^{L/2} \sqrt{1 + 64\dfrac{h^3x^2}{L^4}}dx$

C. $\int_0^{L/2} \sqrt{1 + 64\dfrac{h^2x^2}{L^4}}dx$ **D.** $2\int_0^{L/2} \sqrt{1 + 64\dfrac{h^2x^2}{L^4}}dx$

Q2.02.1 MCQ 2002 1M

The following function has local minima at which value of x, $f(x) = x\sqrt{5 - x^2}$

A. $\frac{-\sqrt{5}}{2}$ **B.** $\sqrt{5}$ **C.** $\sqrt{\frac{5}{2}}$ **D.** $-\sqrt{\frac{5}{2}}$

Q2.02.2 MCQ 2002 1M

Limit of the following sequence as $n \to \infty$ is _____

$x_n = n^{\frac{1}{n}}$

A. 0 **B.** 1 **C.** ∞ **D.** $-\infty$

Q2.01.3 MCQ 2001 1M

Limit of the following series as x approaches $\frac{\pi}{2}$ is

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$$

A. $\frac{2\pi}{3}$ **B.** $\frac{\pi}{2}$ **C.** $\frac{\pi}{3}$ **D.** 1

Q2.00.1 MCQ 2000 1M

The Taylor series expansion of $\sin x$ about $x = \frac{\pi}{6}$ is given by

A. $\frac{1}{2} + \frac{\sqrt{3}}{2}\left(x - \frac{\pi}{6}\right) - \frac{1}{4}\left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12}\left(x - \frac{\pi}{6}\right)^3 + \ldots$

B. $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$

C. $\frac{x - \frac{\pi}{6}}{1!} - \frac{\left(x - \frac{\pi}{6}\right)^3}{3!} + \frac{\left(x - \frac{\pi}{6}\right)^5}{5!} - \frac{\left(x - \frac{\pi}{6}\right)^7}{7!} + \ldots$

D. $\frac{1}{2}$

Q2.99.1 MCQ 1999 1M

Number of inflection points for the curve $y = x + 2x^4$ is _____.

A. 3 **B.** 1 **C.** 0 **D.** 2

Chemical Engineering

Q2.26.1 MCQ 2026 1M

Which of the following is the value of $\lim_{x\to 0} \dfrac{e^x - x - 1}{\cos x - 1}$

A. -1 **B.** 0 **C.** 1 **D.** ∞

Q2.24.1 MCQ 2024 1M

The first non-zero term in the Taylor series expansion of $(1 - x) - e^{-x}$ about $x = 0$ is

A. 1 **B.** -1 **C.** $\frac{x^2}{2}$ **D.** $-\frac{x^2}{2}$

Q2.23.1 MCQ 2023 1M

Which one of the following is the CORRECT value of y, as defined by the expression given below?

$$y = \lim_{x \to 0} \frac{2x}{e^x - 1}$$

A. 1 **B.** 2 **C.** 0 **D.** ∞

Q2.22.1 MCQ 2022 1M

Let $f(x) = e^{-|x|}$, where x is real. The value of $\frac{df}{dx}$ at $x = -1$ is

A. $-e$ **B.** e **C.** $1/e$ **D.** $-1/e$

Q2.22.2 MCQ 2022 1M

The value of the real variable $x \geq 0$, which maximizes the function $f(x) = x^e e^{-x}$ is

A. e **B.** 0 **C.** $1/e$ **D.** 1

Q2.22.3 MCQ 2022 2M

The partial differential equation

$$\frac{\partial u}{\partial t} = \frac{1}{\pi^2} \frac{\partial^2 u}{\partial x^2}$$

where, $t \geq 0$ and $x \in [0, 1]$, is subjected to the following initial and boundary conditions:

$$u(x, 0) = \sin(\pi x)$$

$$u(0, t) = 0$$

$$u(1, t) = 0$$

The value of t at which $\frac{u(0.5, t)}{u(0.5, 0)} = \frac{1}{e}$ is

A. 1 **B.** e **C.** π **D.** $1/e$

Q2.21.1 MCQ 2021 1M

The function $\cos(x)$ is approximated using Taylor series around $x = 0$ as $\cos(x) \approx 1 + ax + bx^2 + cx^3 + dx^4$. The values of a, b, c and d are

A. $a = 1,\ b = -0.5,\ c = -1,\ d = -0.25$

B. $a = 0,\ b = -0.5,\ c = 0,\ d = 0.042$

C. $a = 0,\ b = 0.5,\ c = 0,\ d = 0.042$

D. $a = -0.5,\ b = 0,\ c = 0.042,\ d = 0$

Q2.21.2 MSQ 2021 1M

For the function $f(x) = \begin{cases} -x, & x < 0 \\ x^2, & x \geq 0 \end{cases}$ the CORRECT statement(s) is/are

A. $f(x)$ is continuous at $x = 1$

B. $f(x)$ is differentiable at $x = 1$

C. $f(x)$ is continuous at $x = 0$

D. $f(x)$ is differentiable at $x = 0$

Q2.20.1 MCQ 2020 1M

Consider the hyperbolic functions in **Group – 1** and their definitions in **Group – 2**.

Group – 1		Group – 2	
P	$\tanh x$	**I**	$\frac{e^x + e^{-x}}{e^x - e^{-x}}$
Q	$\coth x$	**II**	$\frac{2}{e^x + e^{-x}}$
R	$\text{sech}\, x$	**III**	$\frac{2}{e^x - e^{-x}}$
S	$\text{cosech}\, x$	**IV**	$\frac{e^x - e^{-x}}{e^x + e^{-x}}$

The correct combination is

A. P – IV, Q – I, R – III, S – II

B. P – II, Q – III, R – I, S – IV

C. P – IV, Q – I, R – II, S – III

D. P – I, Q – II, R – IV, S – III

Q2.20.2 MCQ 2020 2M

The maximum value of the function $f(x) = -\frac{5}{3}x^3 + 10x^2 - 15x + 16$ in the interval $(0.5, 3.5)$ is

A. 0 **B.** 8 **C.** 16 **D.** 48

Q2.19.1 MCQ 2019 1M

The value of the expression $\lim_{x \to \frac{\pi}{2}} \left| \frac{\tan x}{x} \right|$ is

A. ∞ **B.** 0 **C.** 1 **D.** -1

Q2.18.1 NAT 2018 2M

If $y = e^{-x^2}$ then the value of $\lim_{x\to\infty} \frac{1}{x}\frac{dy}{dx}$ is _____

Q2.17.1 NAT 2017 1M

The value of $\lim_{x\to 0} \frac{\tan(x)}{x}$ is _____.

Q2.16.1 NAT 2016 2M

The Lagrange mean-value theorem is satisfied for $f(x) = x^3 + 5$, in the interval $(1, 4)$ at a value (rounded off to the second decimal place) of x equal to _____

Q2.14.4 MCQ 2014 1M

If $f(x)$ is a real and continuous function of x, the Taylor series expansion of $f(x)$ about its minima will **NEVER** have a term containing

A. first derivative

B. second derivative

C. third derivative

D. any higher derivative

Q2.13.1 MCQ 2013 1M

Evaluate

$$\int \frac{dx}{e^x - 1}$$

(Note: C is a constant of integration.)

A. $\frac{e^x}{e^x - 1} + C$ **B.** $\frac{\ln(e^x - 1)}{e^x} + C$

C. $\ln\left(\frac{e^x}{e^x - 1}\right) + C$ **D.** $\ln(1 - e^{-x}) + C$

Q2.12.1 MCQ 2012 2M

If a is a constant, then value of the integral $a^2 \int_0^\infty x e^{-ax} dx$

A. $1/a$ **B.** a **C.** 1 **D.** 0

Q2.11.1 MCQ 2011 1M

Which **ONE** of the following functions $y(x)$ has the slope of its tangent equal to $\frac{ax}{y}$?

A. $y = \frac{x + b}{a}$ **B.** $y = ax + b$

C. $y = \sqrt{\frac{x^2 + b}{a}}$ **D.** $y = \sqrt{ax^2 + b}$

Q2.11.2 MCQ 2011 2M

The value of the improper integral $\int_{-\infty}^{\infty} \frac{dx}{(1 + x^2)}$ is

A. -2π **B.** 0 **C.** π **D.** 2π

Q2.10.1 MCQ 2010 2M

For a function $g(x)$, if $g(0) = 0$ and $g'(0) = 2$, then

$$\lim_{x\to 0} \int_0^{g(x)} \frac{2t}{x} dt$$

is equal to

A. ∞ **B.** 2 **C.** 0 **D.** $-\infty$

Q2.09.1 MCQ 2009 2M

The value of the limit

$$\lim_{x\to \pi/2} \frac{\cos x}{(x - \pi/2)^3}$$

is

A. $-\infty$ **B.** 0 **C.** 1 **D.** ∞

Q2.08.1 MCQ 2008 1M

The limit of $\frac{\sin x}{x}$ as $x \to \infty$ is

A. -1 **B.** 0 **C.** 1 **D.** ∞

Q2.08.2 MCQ 2008 2M

The first four terms of the Taylor series expansion of $\cos x$ about the point $x = 0$ are

A. $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$ **B.** $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!}$

C. $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}$ **D.** $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$

Q2.07.1 MCQ 2007 2M

The family of curves that is orthogonal to

$$xy = c$$

is

A. $y = c_1 x$ **B.** $y = c_1/x$

C. $y^2 + x^2 = c_1$ **D.** $y^2 - x^2 = c_1$

Q2.06.1 MCQ 2006 1M

The derivative of $|x|$ with respect to x when $x \neq 0$ is

A. $|x|/x$ **B.** -1 **C.** 1 **D.** Undefined

Q2.05.1 MCQ 2005 2M

In the domain $-\infty < x < \infty$, the function $y(x) = x^3 e^{-x}$ has

A. no maximum and no minimum

B. one maximum and no minimum

C. no maximum and one minimum

D. one maximum and one minimum

Q2.05.2 MCQ 2005 2M

In the limit $x \to 0$, what is the limiting value of the function $F(x)$ given below?

$$F(x) = \frac{1 - \cos(2x)}{x^2}$$

A. 0 **B.** 1 **C.** 2 **D.** ∞

Q2.04.1 MCQ 2004 1M

If $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$, then $\frac{dy}{dx}$ will be equal to

A. $\sin\left(\frac{\theta}{2}\right)$ **B.** $\cos\left(\frac{\theta}{2}\right)$ **C.** $\tan\left(\frac{\theta}{2}\right)$ **D.** $\cot\left(\frac{\theta}{2}\right)$

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Q2.26.1 MCQ 2026 1M

Which ONE of the following options is the closest approximation to the value of the given definite integral?

$$\int_0^1 e^{-x^2}\, dx$$

Note: Use numerical integration

A. 1 **B.** $\frac{1}{e}$

C. $\frac{e+1}{2e}$ **D.** $\frac{1+e}{2}$

Q2.24.1 NAT 2024 1M

If

$$\lim_{x \to 1}\left(\frac{x^2 - 2ax + b}{x - 1}\right) = 8$$

then $(a - b)$ is _____. (Answer in integer)

Q2.23.1 MCQ 2023 2M

Given, $z(x, y) = e^{x-2y}$, where $x(t) = e^t$ and $y(t) = e^{-t}$. All the variables are real. The total differential $\frac{dz}{dt}$ is

A. $-z(x + 2y)$ **B.** $-z(x - 2y)$

C. $z(x + 2y)$ **D.** $z(x - 2y)$

Q2.23.2 MCQ 2023 1M

The power series expansion of a function is given

$$\frac{1}{x}\ln(1 + x) = 1 + bx + cx^2 + \ldots$$

for $0 < x \leq 1$.

The values of constants b and c, respectively, are

A. $-\frac{1}{2}$ and $\frac{1}{3}$ **B.** $\frac{1}{2}$ and $-\frac{1}{3}$

C. -1 and $\frac{1}{2}$ **D.** 1 and $-\frac{1}{2}$

Q2.23.3 MCQ 2023 1M

What is $\lim_{x \to 0} f(x)$, where $f(x) = x \sin\frac{1}{x}$?

A. 0 **B.** 1

C. ∞ **D.** Limit does not exist

Q2.23.4 NAT 2023 1M

Consider the real-valued function $g(x) = \max\{(x-2)^2, -2x+7\}$, where $x \in (-\infty, \infty)$. The minimum value attained by $g(x)$ is _____ (rounded off to one decimal place).

Q2.22.1 NAT 2022 2M

The value of $\lim_{x \to 1} \frac{x^3 - 3x + 2}{x^3 - x^2 - x + 1}$ is _____.

Q2.21.1 NAT 2021 2M

The minimum value of function f defined by $f(x, y, z) = x^2 + 5y^2 + 5z^2 - 4x + 40y - 40z + 300$ is _____. [in integer]

Q2.21.2 NAT 2021 2M

If $(3i + 1)x + (4i + 4)y + 5 = 0$ with x, y being real and $i = \sqrt{-1}$, then $x =$ _____.
[correct upto one decimal place]

Q2.19.1 MCQ 2019 1M

The solution of $\int_1^a \int_1^b \frac{dxdy}{xy}$ is

A. $\ln(ab)$ **B.** $\ln(a/b)$
C. $\ln(a) + \ln(b)$ **D.** $\ln(a)\ln(b)$

Q2.18.1 MCQ 2018 1M

A real-valued function y of real variable x is such that $y = 5|x|$. At $x = 0$, the function is

A. discontinuous but differentiable
B. both continuous and differentiable
C. discontinuous and not differentiable
D. continuous but not differentiable

Q2.16.1 MCQ 2016 1M

At $x = 0$, the function is

$$f(x) = \left|\sin\frac{2\pi x}{L}\right| (-\infty < x < \infty, L > 0)$$

A. continuous and differentiable.
B. not continuous and not differentiable.
C. not continuous but differentiable.
D. continuous but not differentiable.

Q2.16.2 MCQ 2016 1M

For the two functions

$$f(x, y) = x^3 - 3xy^2 \text{ and } g(x, y) = 3x^2y - y^3$$

Which one of the following options is correct?

A. $\frac{\partial f}{\partial x} = \frac{\partial g}{\partial x}$ **B.** $\frac{\partial f}{\partial x} = -\frac{\partial g}{\partial y}$
C. $\frac{\partial f}{\partial y} = -\frac{\partial g}{\partial x}$ **D.** $\frac{\partial f}{\partial y} = \frac{\partial g}{\partial x}$

Q2.16.3 MCQ 2016 2M

The range of values of k for which the function $f(x) = (k^2 - 4)x^2 + 6x^3 + 8x^4$ has a local maxima at point $x = 0$ is

A. $k < -2$ or $k > 2$ **B.** $k \le -2$ or $k \ge 2$
C. $-2 < k < 2$ **D.** $-2 \le k \le 2$

Q2.16.4 NAT 2016 2M

$\lim_{x\to 0}\left(\frac{e^{5x} - 1}{x}\right)^2$ is equal to _____.

Q2.15.1 MCQ 2015 1M

The function $f(x) = x^2 = x + x + x + \ldots x$ times, is defined

A. at all real values of x
B. only at positive integer values of x
C. only at negative integer values of x
D. only at rational values of x

Q2.15.2 MCQ 2015 2M

The value of $\lim_{(x,y)\to(0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$ is

A. 0 **B.** 1 **C.** $\frac{1}{2}$ **D.** ∞

Q2.15.3 MCQ 2015 2M

The curve $y = x^4$ is

A. concave up for all values of x
B. concave down for all values of x
C. concave up only for positive values of x
D. concave up only for negative values of x

Q2.10.1 MCQ 2010 1M

If $f(x) = \sin|x|$ then the value of $\frac{df}{dx}$ at $x = \frac{-\pi}{4}$ is

A. 0 **B.** $-\frac{1}{\sqrt{2}}$ **C.** $\frac{1}{\sqrt{2}}$ **D.** 1

Q2.10.2 MCQ 2010 1M

The integral $\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{\frac{-x^2}{2}}\,dx$ is equal to

A. $\frac{1}{2}$ **B.** 1 **C.** $\frac{1}{\sqrt{2}}$ **D.** ∞

Q2.09.1 MCQ 2009 1M

The total derivative of the function 'xy' is

A. $x\,dy + y\,dx$ **B.** $x\,dx + y\,dy$
C. $dx + dy$ **D.** $dx\,dy$

Q2.08.1 MCQ 2008 1M

The value of the integral $\int_{-\pi/2}^{\pi/2} (x \cos x)\, dx$ is

A. 0 **B.** π

C. $\pi - 2$ **D.** $\pi + 2$

Q2.08.2 MCQ 2008 1M

The value of the expression $\lim_{x \to 0} \left[\frac{\sin(x)}{e^x x} \right]$ is

A. 0 **B.** 1

C. $\frac{1}{2}$ **D.** $\frac{1}{1+e}$

Q2.07.1 MCQ 2007 1M

What is the value of $\lim_{x \to \pi/4} \frac{\cos x - \sin x}{x - \pi/4}$

A. $\sqrt{2}$ **B.** $-\sqrt{2}$

C. 0 **D.** Limit does not exist

Q2.07.2 MCQ 2007 1M

For the function $f(x, y) = x^2 - y^2$ defined on $\mathbb{R}^2$, the point $(0, 0)$ is

A. a local minimum

B. Neither a local minimum (nor) a local maximum

C. a local maximum

D. Both a local minimum and a local maximum

Q2.95.1 NAT 1995 1M

Given $y = \int_1^{x^2} \cos t\, dt$, then $\frac{dy}{dx} =$ _____.

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

ELECTRONICS & COMMUNICATION ENGINEERING					
Q2.26.1 0.67	Q2.26.2 A	Q2.25.1 A,B	Q2.25.2 B	Q2.25.3 A	Q2.23.1 C
Q2.22.1 B	Q2.22.2 512	Q2.20.1 2.25	Q2.20.2 B	Q2.19.1 2	Q2.19.2 C
Q2.18.1 D	Q2.18.2 0	Q2.18.3 4.50	Q2.17.1 -100	Q2.17.2 C	Q2.17.3 C
Q2.17.4 0.79	Q2.16.1 20	Q2.16.2 B	Q2.16.3 4.66 to 4.76	Q2.16.4 B	Q2.16.5 2
Q2.16.6 C	Q2.16.7 10	Q2.15.1 1	Q2.15.2 3	Q2.15.3 B	Q2.15.4 2
Q2.15.5 B	Q2.15.6 A	Q2.14.1 A	Q2.14.2 864	Q2.14.3 A	Q2.14.4 C
Q2.14.5 0	Q2.14.6 D	Q2.14.7 6	Q2.14.8 C	Q2.14.9 C	Q2.12.1 C
Q2.10.1 A	Q2.09.1 D	Q2.08.1 A	Q2.08.2 A	Q2.08.3 B	Q2.08.4 A
Q2.07.1 A	Q2.07.2 B	Q2.07.3 A	Q2.07.4 C	Q2.07.5 A	Q2.05.1 A
ELECTRICAL ENGINEERING					
Q2.26.1 0.25	Q2.25.1 B	Q2.22.1 A	Q2.21.1 1	Q2.21.2 A	Q2.20.1 D
Q2.20.2 C	Q2.18.1 D	Q2.18.2 12	Q2.18.3 0.50	Q2.17.1 1	Q2.17.2 B
Q2.17.3 5.732	Q2.17.4 40	Q2.17.5 7	Q2.17.6 A	Q2.16.1 0.2928	Q2.16.2 D
Q2.16.3 0	Q2.14.1 A	Q2.14.2 C	Q2.14.3 B	Q2.14.4 B	Q2.14.5 2
Q2.13.1 B	Q2.11.1 D	Q2.11.2 C	Q2.10.1 B	Q2.10.2 B	Q2.09.1 A
Q2.07.1 B	Q2.07.2 D	Q2.06.1 D	Q2.05.1 C	Q2.05.2 A	
INSTRUMENTATION ENGINEERING					
Q2.26.1 D	Q2.26.2 A,D	Q2.25.1 C	Q2.23.1 A	Q2.23.2 1.0	Q2.22.1 -3.0
Q2.21.1 100.0	Q2.21.2 A	Q2.20.1 A	Q2.20.2 0.5	Q2.19.1 B	Q2.18.1 1.0
Q2.18.2 D	Q2.18.3 B	Q2.16.1 3.0	Q2.16.2 0.5	Q2.16.3 -13.0	Q2.15.1 C

Q2.14.1 D	Q2.13.1 A	Q2.11.1 B	Q2.08.1 C	Q2.08.2 C	Q2.08.3 B
Q2.08.4 C	Q2.07.1 D	Q2.07.2 C	Q2.07.3 C	Q2.05.1 C	Q2.05.2 D
Q2.01.1 D	Q2.99.1 C				
COMPUTER SCIENCE & IT					
Q2.26.1 A,C,D	Q2.26.2 A,C	Q2.25.1 A	Q2.24.1 D	Q2.24.2 B	Q2.23.1 B,D
Q2.23.2 0	Q2.22.1 -0.5	Q2.21.1 0.25	Q2.21.2 19	Q2.20.1 A	Q2.19.1 C
Q2.18.1 D	Q2.18.2 0.29	Q2.17.1 C	Q2.17.2 B	Q2.17.3 C	Q2.17.4 8
Q2.17.5 C	Q2.16.1 1	Q2.16.2 9	Q2.15.1 A	Q2.15.2 C	Q2.15.3 0.99
Q2.15.4 -1	Q2.15.5 A	Q2.14.1 4	Q2.14.2 A	Q2.14.3 A	Q2.13.1 A
Q2.12.1 B	Q2.11.1 D	Q2.10.1 B	Q2.09.1 D	Q2.05.1 A	Q2.05.2 A
Q2.04.1 A	Q2.03.1 B				
MECHANICAL ENGINEERING					
Q2.26.1 A	Q2.25.1 A	Q2.23.1 B	Q2.23.2 8	Q2.22.1 A	Q2.22.2 B
Q2.21.1 B	Q2.21.2 D	Q2.21.3 D	Q2.20.1 D	Q2.20.2 D	Q2.19.1 D
Q2.19.2 2.097	Q2.19.3 B	Q2.18.1 A	Q2.17.1 D	Q2.17.2 C	Q2.16.1 -5
Q2.16.2 C	Q2.16.3 C	Q2.16.4 C	Q2.15.1 -0.333	Q2.15.2 C	Q2.15.3 1.8621
Q2.15.4 2	Q2.15.5 A	Q2.14.1 A	Q2.14.2 B	Q2.14.3 D	Q2.14.4 B
Q2.14.5 B	Q2.13.1 C	Q2.12.1 D	Q2.12.2 B	Q2.11.1 D	Q2.10.1 B
Q2.10.2 D	Q2.10.3 C	Q2.10.4 D	Q2.09.1 A	Q2.08.1 A	Q2.08.2 C
Q2.08.3 B	Q2.08.4 D	Q2.08.5 D	Q2.08.6 C	Q2.07.1 B	Q2.07.2 B
CIVIL ENGINEERING					
Q2.26.1 C	Q2.25.1 A	Q2.25.2 B	Q2.25.3 A,D	Q2.25.4 3	Q2.24.1 A

Q2.24.2 B	Q2.24.3 5.13	Q2.23.1 D	Q2.23.2 A	Q2.22.1 A, B	Q2.22.2 *
Q2.22.3 B	Q2.21.1 15	Q2.21.2 0.5	Q2.21.3 0.0	Q2.21.4 C	Q2.20.1 D
Q2.20.2 D	Q2.20.3 D	Q2.19.1 A	Q2.19.2 A	Q2.19.3 D	Q2.19.4 A
Q2.19.5 A	Q2.18.1 D	Q2.18.2 B	Q2.17.1 A	Q2.17.2 A	Q2.17.3 -1
Q2.17.4 B	Q2.17.5 C	Q2.16.1 B	Q2.16.2 B	Q2.16.3 B	Q2.16.4 D
Q2.16.5 D	Q2.16.6 D	Q2.16.7 85.33	Q2.15.1 D	Q2.15.2 C	Q2.15.3 D
Q2.14.1 B	Q2.14.2 A	Q2.14.3 A	Q2.13.1 C	Q2.12.1 C	Q2.11.1 C
Q2.11.2 B	Q2.10.1 A	Q2.10.2 A	Q2.10.3 D	Q2.02.1 D	Q2.02.2 B
Q2.01.3 D	Q2.00.1 A	Q2.99.1 C			
CHEMICAL ENGINEERING					
Q2.26.1 A	Q2.24.1 D	Q2.23.1 B	Q2.22.1 C	Q2.22.2 A	Q2.22.3 A
Q2.21.1 B	Q2.21.2 A,B,C	Q2.20.1 C	Q2.20.2 C	Q2.19.1 A	Q2.18.1 0
Q2.17.1 1	Q2.16.1 2.65	Q2.14.4 A	Q2.13.1 D	Q2.12.1 C	Q2.11.1 D
Q2.11.2 C	Q2.10.1 C	Q2.09.1 A	Q2.08.1 B	Q2.08.2 C	Q2.07.1 D
Q2.06.1 A	Q2.05.1 B	Q2.05.2 C	Q2.04.1 C		
PRODUCTION & INDUSTRIAL ENGINEERING					
Q2.26.1 C	Q2.24.1 4	Q2.23.1 C	Q2.23.2 A	Q2.23.3 A	Q2.23.4 1
Q2.22.1 1.4 to 1.6	Q2.21.1 136	Q2.21.2 2.5	Q2.19.1 D	Q2.18.1 D	Q2.16.1 D
Q2.16.2 C	Q2.16.3 C	Q2.16.4 25	Q2.15.1 B	Q2.15.2 A	Q2.15.3 A
Q2.10.1 B	Q2.10.2 B	Q2.09.1 A	Q2.08.1 A	Q2.08.2 B	Q2.07.1 B
Q2.07.2 B	Q2.95.1 $2x\cos(x^2)$				

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